Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

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Abstract

This dissertation presents CLARION, an explainable (XAI-first) financial-alert system for retail investors in the US equity market (NYSE and NASDAQ). The system raises Telegram alerts from two independent triggers — abrupt market movements, detected as statistical anomalies, and incoming financial news — and, for every alert, exposes a fully traceable explanation: the detected event, the reasoning applied, the sources, and historical precedents. Its core is a news–market correlation engine that, given a news item, retrieves analogous historical news from a knowledge base and measures the impact those precedents had on prices, in the manner of an event study and without lookahead. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components — sentence-embedding retrieval, statistical anomaly detection and transparent explanation — in a functional, explainable and reproducible system. On real data, semantic retrieval clearly outperformed lexical and trivial baselines (cross-ticker precision@5 of 0.51 against 0.24 for random retrieval), and the volatility-normalised detector fired far more consistently across stocks than a fixed threshold. The results are honest and reproducible, with limitations and future work stated explicitly.

Keywords: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Resumo

Esta dissertação apresenta o CLARION, um sistema de alertas financeiros explicável (XAI-first) para investidores de retalho no mercado acionista dos Estados Unidos (NYSE e NASDAQ). O sistema envia alertas via Telegram a partir de dois gatilhos independentes — movimentos abruptos de mercado, detetados como anomalias estatísticas, e novas notícias financeiras — e, para cada alerta, expõe uma explicação totalmente rastreável: o evento detetado, o raciocínio aplicado, as fontes e os precedentes históricos. O seu núcleo é um motor de correlação notícia–mercado que, dada uma notícia, recupera notícias históricas análogas de uma base de conhecimento e mede o impacto que esses precedentes tiveram nos preços, à maneira de um estudo de evento e sem lookahead. Enquanto contribuição de Engenharia de Inteligência Artificial, o trabalho integra, aplica e avalia criticamente componentes existentes — recuperação por embeddings de frases, deteção estatística de anomalias e explicação transparente — num sistema funcional, explicável e reprodutível. Em dados reais, a recuperação semântica superou claramente as baselines lexical e triviais (precision@5 cross-ticker de 0,51 face a 0,24 do acaso) e o detetor normalizado pela volatilidade disparou de forma muito mais consistente entre ações do que um limiar fixo. Os resultados são honestos e reproduzíveis, com limitações e trabalho futuro explícitos.

Palavras-chave: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

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r_t	log-return of an asset on day t	—
μ_t	rolling mean of returns	—
σ_t	rolling standard deviation of returns	—
z_t	z-score of the return on day t	—
k	anomaly threshold (in standard deviations)	—
w	rolling-window length	d

List of Acronyms

AI	Artificial Intelligence.
NASDAQ	National Association of Securities Dealers Automated Quotations.
NYSE	New York Stock Exchange.
RQ	Research Question.

Chapter 1

Introduction

1.1 Contextualization

Most Americans now own stock, either directly or through retirement accounts and mutual funds (Gallup 2025). At the same time, the US equity market has grown into the world's largest and most liquid, reaching US$62.2 trillion at the end of 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025) (Figure 1.1). Trading takes place mainly on the New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations (NASDAQ), and a growing share of it now comes from ordinary people using commission-free mobile apps rather than from professional institutions.

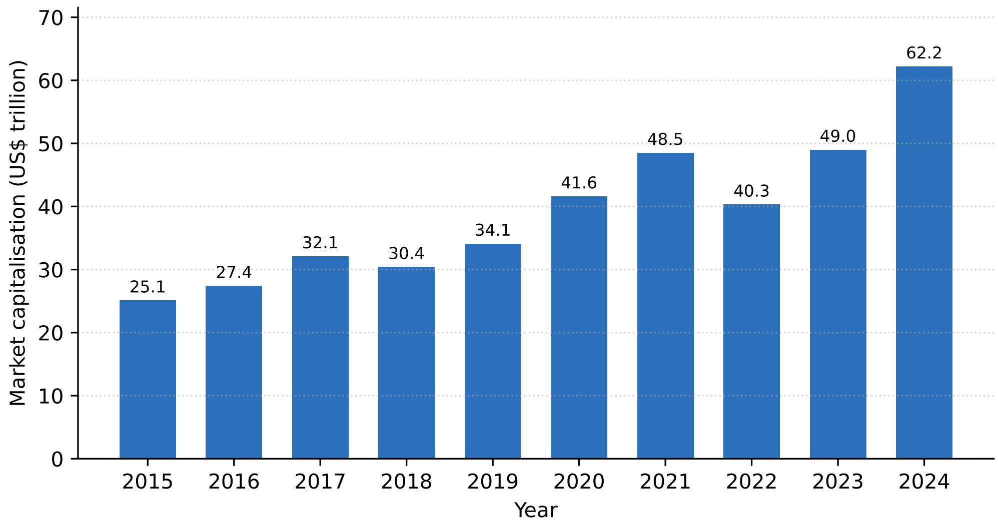

Figure 1.1: US equity market capitalisation, 2015–2024 (US$ trillion). Drawn from data reported in the SIFMA 2025 Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

These non-professional, or "retail", investors are the audience of this work. Stock ownership among them is wide but uneven: it rises to 87% of households earning above US$100,000 and falls to 28% below US$50,000 (Gallup 2025). What they share is a hard position. Markets move continuously across thousands of securities and relevant news arrives throughout the trading day, yet retail investors have none of the tools that banks and funds take for granted.

Artificial intelligence has, meanwhile, spread quickly through finance. A 2026 survey found that 81% of financial firms were adopting Artificial Intelligence (AI) in some form and 71% were already using generative AI (Cambridge Centre for Alternative Finance 2026). Much of this technology is opaque: the strongest models rarely reveal how they reach a conclusion (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). For someone who has to act on an alert and live with the result, that opacity is a real obstacle. Letting the investor see *why* an alert was raised is what this dissertation sets out to do.

1.2 Problem Statement

A retail investor who wants to stay informed faces two distinct challenges. The first is telling a genuinely unusual price move apart from ordinary daily volatility, which takes statistical judgement and constant attention. The second is reading what a news item might mean, which takes knowledge of how similar events played out in the past, knowledge most people do not have to hand. Existing consumer applications often provide alerts with little explanation, or rely on models whose reasoning is hidden from the user.

This work does not try to predict prices. It tries to *explain*. For each abrupt move or relevant news item, the system shows what it detected, how it reached that conclusion, where the information came from, and which past events resemble the current one, alongside what happened to prices afterwards (Figure 1.2).

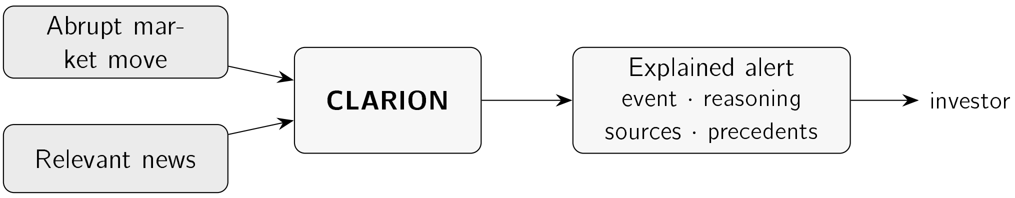

Figure 1.2: In one picture: two triggers — a sudden market move or a relevant news item — feed CLARION, which replies not with a bare notification but with an explained alert that the investor can follow and check.

1.3 Research Questions and Objectives

The aim is to design, build and evaluate an explainable financial-alert system for US retail investors, driven by two independent triggers, a sudden market move and an incoming news item, and delivered over Telegram. Three research questions guide the work:

- **Research Question (RQ)1.** Can transparent statistical methods detect abrupt market movements consistently enough to be useful to a retail investor, while staying explainable?
- **RQ2.** Can a news–market correlation engine retrieve genuinely analogous historical news, and quantify its observed price impact without lookahead, better than trivial baselines?
- **RQ3.** Can the resulting explanations be made faithful to what the system actually computes, and useful to a retail investor?

These questions translate into four objectives: to develop an explainable alerting pipeline driven by the two triggers; to evaluate the anomaly detector against transparent baselines; to evaluate the retrieval of historical news against trivial baselines; and to assess the faithfulness and usefulness of the explanations. The incremental, simplicity-first way the system is built is a methodological choice, described in Chapter 3.

1.4 Scientific Contributions

This is a dissertation in Artificial Intelligence *Engineering*. Its contribution is not a new algorithm but the disciplined assembly of existing ones into a system that works, explains itself, and can be reproduced. Three contributions stand out:

- A reproducible method for relating news to market impact, pairing sentence-embedding retrieval with event-study windows, where the similarity measure and the time horizons are stated explicitly and no information about the future is allowed to leak in.
- **CLARION**, an alerting system that exposes its complete reasoning chain to the user: the detected event, the supporting evidence, the sources, and the relevant historical precedents. Because each explanation is generated directly from the quantities the system computes, the text cannot diverge from the underlying computation.
- A critical evaluation of each core component on real data, against simple and lexical baselines, with the results, the ablations and the limitations all reported plainly.

1.5 Document Structure

The remainder of this dissertation is organised as follows. Chapter 2 reviews the state of the art behind the work: anomaly detection, explainable AI, financial natural-language processing, event studies, and case-based reasoning. Chapter 3 sets out the methods, the datasets and the evaluation design. Chapter 4 presents CLARION itself, at the level of its architecture and the decisions behind it. Chapter 5 evaluates the system through three case studies on real data, and Chapter 6 returns to the research questions, states what was and was not achieved, and points to future work.

Chapter 2

State of the Art

CLARION draws on several fields, and this chapter reviews them with one purpose: to justify each design choice against the alternatives. It begins with what is known about how retail investors behave and why they are exposed to information overload, since that is what the system responds to. It then works through anomaly detection, explainable AI, human trust in automation, financial text representation, information retrieval, event studies, and case-based reasoning, preferring seminal and recent peer-reviewed work and summarising each area in a comparative table. The recurring finding is that the individual pieces are mature, but an integrated, explainable system aimed at the retail investor is still rare.

2.1 Retail Investing and Information Overload

The audience of this work is the non-professional, or "retail", investor. How such investors consume information and react to news is the natural place to start: it both motivates the system and sets what a useful alert must contain.

Behavioural finance documents that real investors deviate systematically from the fully-rational ideal; the survey of Barberis and Thaler (2003) catalogues these deviations and their two pillars, limits to arbitrage and human psychology. A foundational element of that psychology is prospect theory (Kahneman and Tversky 1979), whose central insight — *loss aversion*, the asymmetric weighting of losses against equivalent gains — helps explain why investors are especially sensitive to salient, potentially adverse news, and therefore why a timely, well-explained alert is valuable.

A central and durable finding of this literature is that individual investors are strongly influenced by *attention*. In their seminal study, Barber and Odean (2008) show that individual investors are net buyers of attention-grabbing stocks — those in the news, those with unusually high trading volume, and those with extreme one-day returns — whereas they do not face the same search problem when selling, because they tend to sell only what they already own. The explanation offered is economic: faced with thousands of stocks they could potentially buy, investors fall back on the small subset that has captured their attention. This matters directly here. The events that grab a retail investor's attention, a sharp price move or a salient headline, are the two triggers this work is built on, so an alert that adds context to them intervenes exactly where attention-driven decisions are made. Attention can also be measured: Da, Engelberg, and Gao (2011) show that internet search volume is a timely proxy for it, one that captures retail attention in particular, reinforcing the link between salient events and the investors this work serves.

The role of the media in shaping investor behaviour has likewise been quantified. Tetlock (2007) constructs a daily pessimism measure from a popular Wall Street Journal column and finds that high media pessimism predicts downward pressure on prices followed by a reversion to fundamentals, and that unusually high or low pessimism predicts elevated trading volume. The study is doubly relevant here: it establishes that news content carries measurable, market-moving information, and it is an early demonstration that *textual* signals can be extracted and related to market outcomes — the same premise that underlies the news–market correlation engine of this dissertation.

The composition and weight of retail participation have also shifted markedly in recent years, driven by commission-free mobile brokerages. Studying a large sample of users of one such platform, Welch (2022) documents that this new cohort of investors increased its holdings during the March 2020 bear market rather than panicking, and that an aggregated "crowd consensus" portfolio exhibited both reasonable timing and positive performance over the period studied. Whatever the debate over retail investors' sophistication, the empirical picture is one of a large, active and consequential population — consistent with the broad ownership figures reported by Gallup (2025) and the scale of the market documented by Securities Industry and Financial Markets Association (SIFMA) (2025) in Chapter 1.

Taken together, this literature frames the problem the system addresses. Retail investors are attention-constrained, react to news, and now participate in large numbers, yet they must do so under a continuous flood of price movements and headlines that they cannot individually process. The response pursued here is not to predict the market for them, but to reduce the cost of *interpreting* the events that already command their attention, by attaching to each alert a transparent explanation and concrete historical evidence.

2.2 Anomaly Detection in Financial Markets

The first trigger of the system rests on detecting abnormal price movements, which places it within the mature field of anomaly detection. The survey of Chandola, Banerjee, and Kumar (2009) provides the standard reference and taxonomy, distinguishing *point* anomalies (an individual observation that is abnormal with respect to the rest), *contextual* anomalies (abnormal only in a particular context, such as a time of year), and *collective* anomalies (a set of observations that is jointly abnormal), and grouping detection methods into statistical, distance- and density-based, and machine-learning families. A sharp single-day return is naturally treated as a contextual point anomaly: abnormal relative to the asset's own recent behaviour. Figure 2.1 summarises the families relevant to this work.

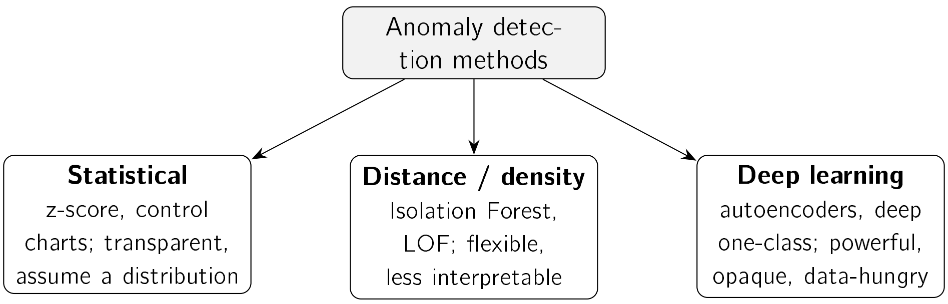

Figure 2.1: Families of anomaly-detection methods relevant to this work, following Chandola, Banerjee, and Kumar (2009) and Pang et al. (2021). Interpretability decreases, and modelling capacity increases, from left to right.

For financial returns, simple statistical methods — flagging deviations from a rolling mean and standard deviation, that is, a z-score — are attractive precisely because they are transparent and easy to justify, an advantage for an explainable system. They do, however, assume a locally stable distribution, an assumption that the rolling estimation window is designed to accommodate. The need for that window is itself well understood in financial econometrics: returns exhibit *volatility clustering* — calm and turbulent periods arrive in runs — formalised by the ARCH model of Engle (1982) and its generalisation GARCH (Bollerslev 1986), which model time-varying variance explicitly. The rolling standard deviation used in this work is, in effect, a deliberately simpler, non-parametric estimate of that same time-varying volatility: it adapts to the current regime without fitting a parametric model, which keeps the detector transparent and free of estimation choices a non-expert could not scrutinise. A full GARCH estimate is a more sophisticated alternative, declined here under the principle of defensible simplicity because the transparent rolling estimate suffices to flag volatility-relative surprise. More expressive unsupervised detectors relax such assumptions: Isolation Forest (F. T. Liu, Ting, and Zhou 2008) isolates anomalies through random partitioning, requiring fewer splits to separate an outlier, and scales well to large, high-dimensional data. The cost is reduced direct interpretability, since the anomaly score is an ensemble property rather than a quantity the user can read off. Density-based detectors such as the Local Outlier Factor (Breunig et al. 2000) take a related route, scoring a point by how isolated it is relative to the density of its neighbourhood; they capture *local* anomalies well but, again, at the expense of a directly readable justification. At the far end of the spectrum, the review by Pang et al. (2021) surveys deep-learning detectors — autoencoders, deep one-class models and others — which can capture rich, non-linear structure but are data-hungry and largely opaque, and whose advantage is most pronounced in high-dimensional settings such as images or complex multivariate streams rather than in a single return series. Within finance specifically, the survey of Ahmed, Mahmood, and Islam (2016) notes that detection is dominated by *unsupervised* methods, precisely because labelled anomalies are scarce — a constraint that also shapes the evaluation strategy adopted in this work, where a label-free consistency argument is given primacy (Chapter 5). Table 2.1 compares the families.

Table 2.1: Anomaly-detection approaches relevant to this work.

Approach	**How it works**	**Strengths / limitations**
Statistical z-score (rolling) (Chandola, Banerjee, and Kumar 2009)	Flags returns that deviate from a rolling mean and standard deviation	Transparent and easy to explain; assumes a locally stable distribution
Isolation Forest (F. T. Liu, Ting, and Zhou 2008)	Isolates points via random partitioning (ensemble)	Scalable, handles high dimensionality; score is not directly interpretable
Deep detectors (Pang et al. 2021)	Learn a model of normality (e.g. autoencoders)	High capacity for complex data; opaque, data-hungry, hard to justify

The trade-off this table makes explicit — capacity against interpretability — is the one that governs the design choice in Chapter 3. Because the system must *explain* every alert to a non-expert, and because the signal is a single, low-dimensional return series rather than a high-dimensional stream, the transparent statistical detector is preferred, with the more expressive families noted as alternatives whose opacity is not warranted by the problem at hand.

2.3 Explainable Artificial Intelligence

As AI systems are deployed in consequential settings, the inability to inspect their reasoning has become a central concern, and explainable AI (XAI) is the field that addresses it. The surveys of Adadi and Berrada (2018) and Barredo Arrieta et al. (2020) map the area: its motivations (trust, accountability, debugging, regulatory compliance), a taxonomy that separates models that are *transparent* (interpretable in their own right) from *post-hoc* methods that explain an already-trained black box, and the open challenges on the path to responsible AI. The earlier survey of Guidotti et al. (2018) organises the post-hoc literature specifically by the problem being solved — explaining a whole model, explaining a single outcome, or inspecting the internals — which is a useful lens for situating the techniques most relevant here. Figure 2.2 sketches this division. A recurring caution in the field is that "interpretability" is not a single, well-defined property: Lipton (2018) argues that the term conflates several distinct desiderata (simulatability, decomposability, post-hoc justification) that serve different stakeholders, so a system should state *which* notion it provides. This work is explicit on that point: it offers transparency of the decision logic and traceable evidence, rather than a contested claim of total interpretability.

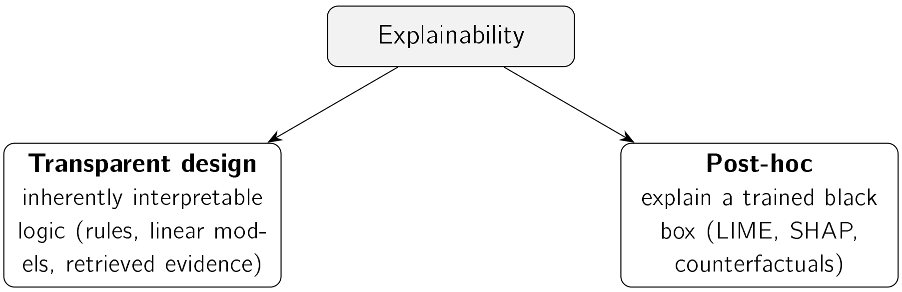

Figure 2.2: The two broad routes to explainability (Barredo Arrieta et al. 2020; Guidotti et al. 2018): building models that are transparent by design, or explaining a black box after the fact. This work follows the former.

Among post-hoc, model-agnostic techniques, two are widely used. LIME (Ribeiro, Singh, and Guestrin 2016) explains a single prediction by fitting a simple, interpretable surrogate model in the local neighbourhood of the instance; it is intuitive and broadly applicable, but its explanations can be unstable under resampling and are only locally faithful. SHAP (Lundberg and S.-I. Lee 2017) attributes a prediction to its input features using Shapley values from cooperative game theory, which gives it desirable consistency properties and a unifying theoretical grounding, at a higher computational cost. Both are valuable when one is committed to a black-box model and needs to interrogate it after the fact.

A strand of the literature questions that commitment directly. Rudin (2019) argues forcefully that, for high-stakes decisions, one should *stop explaining* black-box models and instead use models that are interpretable in the first place, on the grounds that post-hoc explanations are approximations that can mislead and can entrench poor practice. This position is particularly pertinent to a tool that a non-expert will act upon financially: an explanation that is itself an approximation of an opaque model carries a risk that a transparent design avoids by construction. The present work takes this view as a guiding principle, pursuing explainability primarily through transparent design — a statistical detector whose every quantity is exposed, and retrieved historical precedents shown as direct evidence — and treating feature attribution as an optional complement rather than the foundation.

What makes an explanation *good* is, moreover, not purely a technical matter. Miller (2019) draws on the social sciences to argue that human explanations are contrastive, selective and social — people want to know why *this* rather than *that*, and expect a few relevant causes, not an exhaustive account. The alerts in this work are designed in that spirit: each presents a small set of the most relevant precedents and the specific quantities behind the decision, rather than an exhaustive model dump.

Finally, the field has recognised that explanations must themselves be evaluated rather than assumed. Doshi-Velez and Kim (2017) propose a taxonomy for the rigorous evaluation of interpretability, distinguishing application-grounded, human-grounded and functionally-grounded assessments, and emphasising *fidelity* — the degree to which an explanation reflects the system's actual computation — as a core criterion. This framing is adopted in Chapter 5, where the explanation is assessed for faithfulness by construction, with the (human-grounded) question of usefulness stated openly as a limitation. Table 2.2 summarises the approaches considered.

Table 2.2: Explainability approaches considered.

Method	**Type**	**Strengths / limitations**
LIME (Ribeiro, Singh, and Guestrin 2016)	Local surrogate, post-hoc	Intuitive, model-agnostic; explanations can be unstable, only locally faithful
SHAP (Lundberg and S.-I. Lee 2017)	Shapley-value attribution, post-hoc	Consistent, theoretically grounded; computationally costly
Transparent design (Barredo Arrieta et al. 2020; Rudin 2019)	Inherently interpretable logic	No approximation, faithful by construction; constrains model choice

2.4 Trust and Appropriate Reliance in Decision Support

Explainability is, ultimately, in service of a human decision, so the literature on how people rely on automated aids is directly relevant. An alert is only useful if the investor relies on it *appropriately* — neither dismissing a sound warning nor acting blindly on a fallible one. The human-factors account of J. D. Lee and See (2004) frames this precisely: trust is the attitude that an aid will help achieve one's goals under uncertainty, and the design goal is *calibrated* trust — reliance matched to the system's actual competence — rather than maximal trust. Two failure modes follow: *disuse*, ignoring a capable aid, and *misuse*, over-relying on a fallible one. For a tool acted upon with real money, both are costly, which raises the stakes for how alerts are presented.

Explanations are the principal means of calibration. By exposing why an alert fired and what evidence supports it, the system lets the user judge *when* to rely on it, which is the human-side rationale for the transparent-by-design choice of Section 2.3: transparency is not only an ethical or regulatory desideratum but a mechanism for appropriate reliance. Recent empirical work tempers this hope, however. Bansal et al. (2021) show that explanations do not automatically improve human–AI team performance and can even induce *over-reliance*, with users accepting confident explanations of incorrect outputs. The lesson adopted here is concrete: the system explains with *verifiable evidence* — the actual z-score and its parameters, and real retrieved precedents with their measured impact — rather than persuasive narrative, and every alert states plainly that it reports past outcomes, not predictions. These are deliberate choices aimed at appropriate, not maximal, reliance, and they connect the explainability goal of this work to the responsible-AI commitments set out in Chapter 3.

2.5 Financial NLP and Text Representation

The news trigger depends on comparing the meaning of news items, which requires representing text numerically. The literature on financial natural-language processing spans three broad generations: hand-crafted lexicons, learned word embeddings, and contextual neural language models. The survey of Kearney and S. Liu (2014) provides a useful map of the financial-text literature, organising it by information source, content-analysis method and empirical model, and documents the field's steady move from dictionaries towards machine-learned representations — the same trajectory this section follows.

The earliest and still-influential approach in finance is the domain dictionary. Loughran and McDonald (2011) show that general-purpose sentiment word lists misclassify ordinary financial vocabulary — words such as "liability" or "tax" that are negative in everyday language but neutral in a financial report — and develop finance-specific word lists that better capture tone in regulatory filings, relating them to returns, volume and volatility. Lexicon methods are fully transparent and remain a strong baseline, but they are bounded by their vocabulary and cannot capture meaning that is not expressed by listed words.

Learned embeddings address this limitation by placing words in a continuous vector space where proximity reflects distributional similarity. Word2vec (Mikolov et al. 2013) learns such vectors efficiently from local context windows, and GloVe (Pennington, Socher, and Manning 2014) obtains comparable representations from global word–word co-occurrence statistics. Both, however, assign a single vector to each word irrespective of context, so they cannot distinguish the different senses a word takes in different sentences. The Transformer architecture (Vaswani et al. 2017), built entirely on attention mechanisms and dispensing with recurrence, removed this limitation and reshaped the field: BERT (Devlin et al. 2019) uses it to produce deeply *contextual* token representations that substantially advanced language understanding. Using BERT directly to compare two texts is nonetheless expensive, because it processes sentence pairs jointly; Sentence-BERT (Reimers and Gurevych 2019) resolves this by producing a single fixed-length embedding per sentence that can be compared with a cheap cosine similarity — exactly the property on which precedent retrieval in this work relies. Should the knowledge base grow to millions of items, this exact search can be replaced by an approximate nearest-neighbour index such as those of Johnson, Douze, and Jégou (2021) without changing the representation, a scalability path noted here and revisited as future work.

Two further developments are relevant. First, domain adaptation: FinBERT models fine-tune BERT on financial text, for sentiment classification (Araci 2019) and for broader financial communications (Y. Yang, Uy, and Huang 2020), improving on general-purpose models within the domain. Second, the recent rise of large language models, exemplified in finance by BloombergGPT (Wu et al. 2023), a fifty-billion-parameter model trained on a large financial corpus that outperforms comparable general models on financial tasks. Such models are powerful but costly to run and opaque, and they fall outside this work's constraint of free, transparent and reproducible components; they are noted as part of the landscape and as future work rather than as a chosen tool. Table 2.3 compares the representations.

Table 2.3: Text-representation approaches for financial news.

Approach	**Representation**	**Strengths / limitations**
Finance lexicons (Loughran and McDonald 2011)	Curated word lists / tone	Fully transparent; bounded by vocabulary
word2vec / GloVe (Mikolov et al. 2013; Pennington, Socher, and Manning 2014)	Static word vectors	Efficient; one vector per word, no context
BERT (Devlin et al. 2019; Vaswani et al. 2017)	Contextual token embeddings	Strong understanding; pairwise similarity is costly
Sentence-BERT (Reimers and Gurevych 2019)	Sentence embeddings (siamese)	Efficient cosine similarity; the choice in this work
FinBERT (Araci 2019; Y. Yang, Uy, and Huang 2020)	Domain-adapted language model	Finance-tuned; oriented to sentiment / communications
Financial LLMs (Wu et al. 2023)	Large generative model	Very capable; costly, opaque, outside free-tier scope

The progression in Table 2.3 clarifies the design choice. Sentence-BERT occupies the sweet spot for this work: it captures contextual meaning well beyond lexicons and static embeddings, yet, unlike raw BERT or large language models, it yields directly comparable sentence vectors at low cost, which keeps the retrieval step both reproducible and free to run.

2.6 Information Retrieval and Ranking Evaluation

Representing news as vectors (Section 2.5) is only half of the news trigger; the other half is to *rank* a corpus of past news by relevance to a query, which is the defining problem of information retrieval (IR). Grounding the engine and, crucially, its evaluation in this field is what makes the metric used later defensible rather than arbitrary.

The lineage is direct. The vector-space model of Salton, Wong, and C. S. Yang (1975) established the idea that documents and queries can be represented as vectors and ranked by their similarity, classically the cosine of the angle between them. Embedding-based retrieval, as used in this work, is the same ranking paradigm with a richer representation: dense sentence embeddings replace sparse term vectors, but the ranking step — cosine similarity, take the top results — is unchanged. Classical *lexical* ranking refines the term-vector view by weighting terms according to their informativeness, from TF-IDF weighting in the vector-space model to the probabilistic BM25 framework of Robertson and Zaragoza (2009), which remains a strong baseline in modern IR. The well-known weakness of all such lexical methods is the *vocabulary-mismatch* problem: a query and a genuinely relevant document that happen to use different words score as dissimilar. Overcoming this mismatch is precisely the motivation for semantic retrieval, and it is why this work pits dense embeddings against a lexical, term-overlap baseline of exactly this family (Chapter 5).

Because retrieval returns a *ranked list* rather than a single answer, it is evaluated with rank-aware measures. The standard reference (Manning, Raghavan, and Schütze 2008) sets out the common family — precision and recall, mean average precision, and normalised discounted cumulative gain — of which *precision@k*, the fraction of the top k results that are relevant, is the most direct and interpretable. This work adopts precision@k deliberately: an alert can only show the investor a handful of precedents, so what matters is whether the *top few* retrieved cases are relevant, exactly what precision@k measures. Reporting it against random, recency and lexical baselines, as the standard methodology advises, is what turns an isolated number into an interpretable one.

In these terms, the retrieval component of this dissertation is a domain application of mature IR: it reuses an established representation-then-rank pipeline and an established evaluation metric, and applies them to the specific problem of finding analogous financial news. The contribution is that application and the pairing of retrieval with measured market impact, not a new retrieval algorithm — consistent with the engineering framing of the work as a whole.

2.7 Event Studies and News–Market Impact

Having retrieved analogous past news, the system must quantify the impact those events had, which is the province of the *event study*. The methodology originates with Fama et al. (1969), whose study of stock-price adjustment to new information introduced the event-study design and provided an early test of market efficiency by measuring returns around an event date. Brown and Warner (1985) subsequently established the now-standard practice for event studies based on daily returns, characterising how abnormal returns should be measured and tested over short post-event windows — the basis for the +1, +3 and +5-day horizons used in this work. Crucially for the honesty of the present design, the event-study impact is measured strictly *after* the event, so it characterises an outcome rather than forecasting one.

This restraint is not merely cautious; it rests on finance theory. The efficient-market hypothesis of Fama (1970) holds that prices already reflect available information, so that consistently forecasting returns from public news is, by construction, very hard: new information is impounded into prices rapidly, which is precisely why an event study measures a short-window *reaction* rather than a predictable, exploitable signal. Read this way, the theory reframes the design choice of this work — rather than competing with market efficiency by trying to predict prices, the system measures and explains the reactions that have *already* followed comparable news, which is both more honest and more defensible for a non-expert audience.

Relating *news text* to such impact is the subject of a growing literature, of which Tetlock (2007) is an early and influential example, demonstrating a measurable link between media tone and subsequent market behaviour. A more recent strand goes further and attempts to *predict* price movements from news — a body of work surveyed by Xing, Cambria, and

Welsch (2018); Ding et al. (2015), for instance, learn event embeddings from headlines with a deep network to forecast index movements. This dissertation deliberately stops short of prediction — it *measures* the impact that analogous past events had and presents it as evidence, which keeps the system within its no-forecasting scope and avoids the opacity and over-claiming that price-prediction models invite. Conducting such an impact analysis historically and at scale requires news aligned to prices, which is the contribution of the FNSPID dataset (Dong, Fan, and Peng 2024): financial news time-aligned with daily prices for thousands of companies, enabling a news-to-impact knowledge base without bespoke scraping. Its limitations are those of any corpus — coverage and time period — and are documented in the data card of Chapter 3. The combination of a semantic retrieval step with an event-study impact measurement over such a corpus is the methodological core of this dissertation.

2.8 Case-Based Reasoning and Precedent Retrieval

The way this work *uses* retrieved news — as analogous past cases whose outcomes inform the present — corresponds to a well-established artificial-intelligence paradigm: case-based reasoning (CBR). In the foundational account of Aamodt and Plaza (1994), a CBR system solves a new problem by *retrieving* the most similar past cases, *reusing* their solutions, and (optionally) *revising* and *retaining* the outcome. The correlation engine of this dissertation is, in these terms, the retrieve-and-reuse core of a CBR system: each historical news item, paired with its observed price impact, is a *case*; a new headline is the query; semantic similarity performs retrieval; and the precedents' observed impact is reused as evidence.

This framing matters for two reasons. First, it situates the contribution within a recognised paradigm rather than presenting precedent retrieval as ad hoc. Second, CBR is intrinsically aligned with the explainability goal of this work: an explanation of the form "this resembles these specific past cases, which behaved thus" is example-based and immediately intelligible to a non-expert, in contrast to the feature-attribution explanations of opaque models discussed in Section 2.3. The system deliberately stops at retrieve-and-reuse — it presents the precedents as evidence and does not automatically *revise* them into a prediction — which keeps it within its no-forecasting scope.

2.9 Existing Tools for the Retail Investor

The sections above reviewed the academic foundations; it is equally important to ask what the retail investor can already use in practice, since that is the baseline any new system must improve upon. Three broad categories of tool are widely available today, and each addresses a part of the problem this work targets while leaving a characteristic gap.

The first and most ubiquitous category is the *price alert* offered by virtually every brokerage and market-data application. The investor sets a threshold — a target price, or a percentage move — and is notified when it is crossed. Such alerts are immediate, free and simple, and they map naturally onto the attention-driven behaviour documented by Barber and Odean (2008): they fire precisely on the extreme moves that capture attention. Their limitation is equally clear. A fixed threshold is not calibrated to each asset's own volatility, so it fires constantly on volatile stocks and rarely on calm ones — the very inconsistency that the volatility-normalised detector of this work is designed to remove (Chapter 5). More fundamentally, a price alert reports only *that* a level was crossed; it offers no explanation of *why* and no context for judging whether the move is unusual.

The second category comprises *news aggregators and sentiment dashboards*, which collect headlines for a watchlist and, increasingly, attach a sentiment score to each item. These tools build directly on the finding that textual tone carries market-relevant information (Tetlock 2007) and on finance-specific sentiment lexicons (Loughran and McDonald 2011). They are valuable for surfacing *what* is being said, but two gaps remain. A single sentiment polarity (positive / negative) is a thin summary that says little about likely magnitude or about whether comparable news has mattered before; and the scoring is often proprietary and opaque, so the user cannot see why an item was labelled as it was — the opposite of the transparency this work requires. Crucially, such tools rarely connect a current headline to concrete *historical precedents* and their measured outcomes, which is the core of the engine proposed here.

The third category is the *robo-advisor*, the most studied retail-fintech tool of the last decade. D'Acunto, Prabhala, and Rossi (2019) show, on real adopters, that automated advice can deliver genuine benefits — improved diversification and reduced behavioural biases such as the disposition effect and trend chasing — and the systematic review of Cardillo and Chiappini (2024) confirms a mature and fast-growing literature. Robo-advisors, however, operate at a different altitude and with a different philosophy from this work. They *act on the investor's behalf*, allocating and rebalancing a portfolio, whereas the system proposed here neither advises nor trades; it explains discrete events and leaves every decision to the user. And their allocation logic is typically proprietary, so the investor delegates rather than understands — again the reverse of an explanation-first design. Table 2.4 positions the three categories against the system of this dissertation.

Table 2.4: Existing retail tools versus the system proposed in this work.

Tool	**What it does**	**Explains why?**	**Shows precedents?**	**Acts for user?**
Brokerage price alert	Fires on a fixed price / % threshold	No	No	No
News / sentiment app (Tetlock 2007)	Aggregates news, scores sentiment	Partly (opaque score)	No	No
Robo-advisor (D'Acunto, Prabhala, and Rossi 2019)	Allocates and rebalances a portfolio	Rarely (proprietary)	No	Yes
This work (CLARION)	Explains events, shows evidence	Yes, by design	Yes (measured impact)	No

The pattern in Table 2.4 is the practical counterpart of the academic gap identified throughout this chapter. Each existing tool occupies one corner of the problem — alerting, news, or allocation — but none combines volatility-aware detection, transparent explanation, and concrete historical evidence in a single tool that informs rather than decides. That combination is the niche this work sets out to fill.

2.10 Comparative Analysis and Positioning

Across these areas, the literature offers strong individual components but few integrated, explainable, retail-facing systems that combine them. The choices made in this work, and the alternatives against which they were weighed, are summarised in Table 2.5. The guiding principle is *defensible simplicity*, aligned with the interpretable-by-design argument of Rudin (2019): where two options perform comparably, the more transparent and easier-to-explain one is preferred, because the system's purpose is to be understood and acted upon by a non-expert.

Table 2.5: Component choices in this work versus alternatives.

Component	**Chosen**	**Alternative**	**Why**
Anomaly detection	z-score (rolling)	Isolation Forest, deep detectors	Transparency; low-dimensional signal
News retrieval	Sentence-BERT + cosine	Lexicons, word2vec, LLMs	Contextual yet cheap and reproducible
Impact measurement	Event-study windows	—	Standard, reproducible, no forecasting
Explanation	Transparent design + precedents	LIME / SHAP only	Faithful by construction

This positioning also clarifies the nature of the contribution. None of the individual building blocks is novel; what is comparatively scarce, and what this work provides, is their disciplined integration into a single explainable system for the retail investor, with each choice justified against the alternatives reviewed above.

2.11 Chapter Conclusions

This chapter reviewed the foundations of the work across several connected areas. The behavioural-finance literature established that retail investors are attention- and news-driven, motivating an explainable alerting tool rather than a predictive one. The anomaly-detection literature supplied a clear capacity–interpretability trade-off that favours a transparent statistical detector for a low-dimensional return series. The XAI literature, and in particular the interpretable-by-design argument, supported pursuing explainability through transparent design and assessing it by fidelity, while the trust-in-automation literature reframed that transparency as a means to *appropriate reliance* by a non-expert. The financial-NLP literature traced the path from lexicons to sentence embeddings and identified Sentence-BERT as the appropriate, reproducible choice for semantic retrieval, which information-retrieval theory frames as a representation-then-rank task to be judged by precision@k. Event-study methodology and the FNSPID dataset, in turn, provided the means to measure news impact honestly and at scale. Finally, case-based reasoning supplied the paradigm that ties these together — retrieving analogous past cases and reusing their outcomes as transparent evidence. What the literature offers in mature individual components, it largely lacks as an *integrated*, explainable, retail-facing system — the gap this dissertation addresses, and which the following chapters develop.

Chapter 3

Methods and Materials

This chapter sets out how the system was built and how it is judged: the overall approach, the datasets (documented in a reproducible data card), the models behind each component, the responsible-AI considerations, and the evaluation design, which was fixed before any experiment was run. One idea runs through all of it: keep things as simple as they can defensibly be, so that where two options perform about the same, the more transparent one wins. CLARION, the system these methods produce, is described in Chapter 4; the experiments that apply them are reported in Chapter 5.

3.1 Approach and Methodology

The work follows an incremental, end-to-end engineering methodology. Rather than building each component to completion in isolation, a thin slice of the system was first built and proven end to end — from data acquisition through to a delivered alert — so that integration risk was identified and controlled early. Each component was then developed in its simplest defensible form and made more sophisticated only where the added complexity was justified by a measurable or clearly-argued benefit. This reflects the framing of the work as Artificial Intelligence *engineering*: as discussed in Chapter 2, the individual building blocks already exist in the literature, so the contribution lies in their disciplined integration, application and critical evaluation, not in the invention of new algorithms.

Two consequences of this methodology recur throughout the design. First, every component is specified by its *responsibility* and its *interface* rather than by a particular implementation, which keeps the system testable and makes principled substitution possible (for example, exchanging one embedding model for another). Second, the methods chosen are constrained by the project's non-negotiable requirements: only free and reproducible resources, a focus on the US market, transparency at every step, and no price prediction or trading. These requirements act as a filter on the design space before any performance consideration.

3.2 Datasets and Data Sources

The system uses two clearly separated data layers, sharing a common schema so that they are directly comparable: a **historical** layer, built offline, that supplies the precedents, and a **live** layer that supplies real-time prices and news (Figure 3.1). This separation is deliberate: the historical layer must be rich and stable, whereas the live layer must be timely and inexpensive.

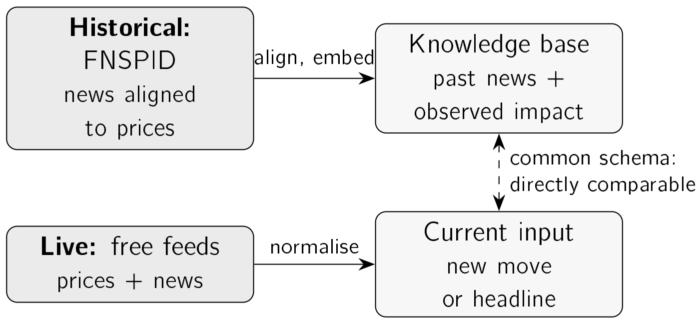

Figure 3.1: The two data layers. The historical layer turns the FNSPID corpus into a knowledge base of past news and its observed price impact; the live layer normalises real-time prices and news into the same fields, so a current event can be compared directly against the precedents.

3.2.1 Historical Layer: the FNSPID Corpus

The intended primary source for the historical layer is the FNSPID dataset (Dong, Fan, and Peng 2024), which provides financial news time-aligned with daily prices for thousands of US companies — precisely the alignment required to relate news to subsequent price impact without bespoke scraping. The dataset's news component is large (on the order of tens of gigabytes), which directly motivates the streaming, filtered ingestion described below. Table 3.1 records the data card: the choices that make the historical subset reproducible.

Table 3.1: Data card for the historical layer (FNSPID).

Item	Value
Source	FNSPID (Dong, Fan, and Peng 2024) (financial news time-aligned with prices)
Licence	Creative Commons BY-SA 4.0 (attribution required)
Fields used	date, ticker, headline (mapped to a common schema)
Tickers (15)	AAPL, MSFT, AMZN, GOOGL, NVDA, TSLA, META, JPM, BAC, XOM, CVX, JNJ, PFE, WMT, KO
Sectors (5)	Technology, banking, energy, health, consumer staples
Time window	2018–2023 (daily granularity)
Prices	Daily closing prices for the same tickers
Governance	Large data not versioned (recreated by script); only small samples committed

The fifteen large-capitalisation tickers were chosen to span five economic sectors, which is what makes the cross-sector evaluation of Chapter 5 meaningful while keeping the corpus tractable on a modest machine; the six-year window (2018–2023, the extent of the dataset) is recent enough to be relevant yet long enough to contain precedents. These choices are documented rather than fixed: a different ticker set or window can be substituted without any change to the methodology.

3.2.2 Preprocessing and Event Alignment

Building the historical knowledge base from the corpus involves three steps. First, the news file is read in a streaming fashion and filtered to the chosen tickers and window, so that only the relevant subset is materialised rather than the full multi-gigabyte corpus. Second, each retained news item is normalised to the common schema (date, ticker, headline). Third, each item is aligned to the market and its impact is measured. The alignment rule is important for honesty: the event day is taken to be the first trading day *on or after* the news date, and the price impact is measured from that day's closing price onward. Measuring from the close, rather than the open, avoids crediting the system with a move already embedded in the opening price, and measuring strictly forward enforces the no-lookahead guarantee of Section 3.6. Items for which no aligned price series is available, or whose forward window extends beyond the available data, are discarded rather than imputed. Algorithm 3.1 collects these steps into the build procedure.

Algorithm 3.1 Knowledge-base construction with event alignment.

Require: news items $\{(d_i, c_i, h_i)\}$; daily close prices P_c per ticker; horizons $H = \{1, 3, 5\}$
Ensure: knowledge base $\mathcal{K}$ of cases $(d, c, h, \mathbf{e}, \text{impact})$

$\mathcal{K} \leftarrow \emptyset$
for each news item (d_i, c_i, h_i) **do**
  $e_i \leftarrow$ first trading day $\geq d_i$ in P_{c_i} ▷ event alignment
  if e_i exists **then**
    $\mathbf{e}_i \leftarrow \text{embed}(h_i)$ ▷ Sentence-BERT
    for $\tau \in H$ **do**
      $\text{impact}_i[\tau] \leftarrow$ return from $\text{close}(e_i)$ to $\text{close}(e_i{+}\tau)$, or n/a if unavailable
    end for
    $\mathcal{K} \leftarrow \mathcal{K} \cup \{(d_i, c_i, h_i, \mathbf{e}_i, \text{impact}_i)\}$
  end if
end for
return $\mathcal{K}$

3.2.3 Live Layer and Evaluation Dataset

The live layer provides, in real time, prices from a free market-data service (with a second free provider as a fallback) and news from a free company-news service and from financial syndication feeds. Free news services are deliberately *not* used to construct the historical layer, because their free tiers are typically restricted to a short, recent window and are delayed; they are, however, entirely adequate as the live trigger and as a source of *real* news for a preliminary evaluation. For the latter purpose, a set of 3,714 real headlines for the fifteen tickers was collected from the free company-news service (the most recent months available), and is used for the retrieval study in Chapter 5. This set is genuinely real but recent, which is why the full multi-year FNSPID build remains the richer source identified as future work.

3.2.4 Data Governance and Attribution

Two governance rules apply to both layers. First, large data and models are never versioned: they are recreated by the ingestion process, and only small illustrative samples are kept under version control. Second, third-party news text is not republished: the dataset is cited and attributed as required by its Creative Commons BY-SA 4.0 licence (Dong, Fan, and Peng 2024), and only minimal headline excerpts appear in this document.

3.3 Models and Techniques

3.3.1 Statistical Anomaly Detection

Abrupt movements are detected with a transparent statistical rule. Let r_t be the (log-)return of an asset on day t, and let μ_t and σ_t be the mean and standard deviation of returns over a trailing window of w days that ends strictly before t. The day is flagged as anomalous when the absolute z-score exceeds a threshold k:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \qquad \text{anomaly} \iff |z_t| > k. \tag{3.1}$$

This belongs to the statistical family of detectors (Chandola, Banerjee, and Kumar 2009) and is chosen for its transparency: the investor can be shown exactly why an alert fired, in terms of how many standard deviations the move represented. As established in Chapter 2, more expressive detectors such as Isolation Forest (F. T. Liu, Ting, and Zhou 2008) or deep models (Pang et al. 2021) exist, but their reduced interpretability is not warranted for a single, low-dimensional return series whose alerts must be explained to a non-expert. The window w and threshold k are explicit design choices, and their effect is examined by ablation in Chapter 5. Algorithm 3.2 states the rule precisely; note that the window for day t uses only days strictly before t, so no future information enters the decision.

Algorithm 3.2 Rolling z-score anomaly detection (no lookahead).

Require: return series $r_1, \ldots, r_T$; window w; threshold k
Ensure: flag a_t for each day, with the quantities behind it

for $t \leftarrow w + 1$ **to** T **do**
  $W \leftarrow (r_{t-w}, \ldots, r_{t-1})$ ▷ strictly earlier days
  $\mu_t \leftarrow \text{mean}(W)$; $\sigma_t \leftarrow \text{std}(W)$
  $z_t \leftarrow (r_t - \mu_t) \,/\, \sigma_t$
  $a_t \leftarrow (|z_t| > k)$
end for
return for each flagged t, the tuple $(z_t, \mu_t, \sigma_t, w, k)$ ▷ exposed to the explanation engine

A small worked example shows why this normalisation matters. Suppose two stocks both fall 3.2% today. The first has been calm lately (recent mean $\mu \approx 0.04\%$, standard deviation $\sigma \approx 0.40\%$); the second is volatile ($\mu \approx 0.10\%$, $\sigma \approx 1.50\%$). The same raw move produces very different z-scores (Table 3.2): for the calm stock $z \approx -8.1$, a clear anomaly, but for the volatile stock $z \approx -2.2$, an ordinary day. A fixed percentage threshold would flag both identically; the z-score does not, which is exactly the behaviour wanted.

Table 3.2: Worked example: the same -3.2% move on a calm and a volatile stock.

Quantity	**Calm stock**	**Volatile stock**
Recent mean μ	0.04%	0.10%
Recent standard deviation σ	0.40%	1.50%
Today's return r	−3.2%	−3.2%
z-score $(r - \mu)/\sigma$	**−8.1**	**−2.2**
Flagged at $k = 3$?	yes	no

3.3.2 Semantic Retrieval and Impact Measurement

The news–market correlation engine is the core of the work and combines two techniques. For retrieval, each news item is encoded into a sentence embedding with a Sentence-BERT model (Reimers and Gurevych 2019), chosen (Section 2.5) because it captures contextual meaning yet produces directly comparable fixed-length vectors, so that the nearest historical items to a query can be found efficiently with cosine similarity. For impact, the cumulative return of the asset is measured over fixed post-event windows (+1, +3 and +5 trading days) in the manner of an event study (Brown and Warner 1985; Fama et al. 1969). Both the similarity metric and the impact windows are explicit, documented choices and together constitute the methodological contribution. Impact is measured strictly after the event, so the evidence characterises an outcome rather than forecasting one. Algorithm 3.3 sets out the retrieval-and-reuse step — the case-based-reasoning core of Section 2.8 — for an incoming headline.

Algorithm 3.3 Precedent retrieval and impact reporting (news trigger).

Require: headline h; knowledge base $\mathcal{K} = \{(d_j, c_j, h_j, \mathbf{e}_j, \text{impact}_j)\}$; neighbours k; horizon τ
Ensure: top-k precedents with similarity and observed impact, and their mean impact

```
q ← embed(h)                               ▷ same Sentence-BERT model as the KB
for each case j ∈ K do
    s_j ← cos(q, e_j)                      ▷ cosine over L2-normalised vectors
end for
S ← indices of the k largest s_j
return {(d_j, c_j, h_j, s_j, impact_j[τ]) : j ∈ S} and mean_{j∈S} impact_j[τ]
```

The impact itself is read off a short window after the event, as Figure 3.2 shows. The news may arrive on a non-trading day, so the *event day* is the first trading day on or after it; the return is then accumulated from that day's close to the close +1, +3 and +5 trading days later. Anchoring on the close, and looking only forward, is what keeps the measure free of lookahead: it records what happened *after* the event became actionable, never before.

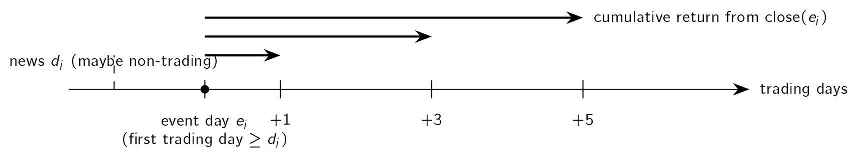

Figure 3.2: How impact is measured. The news date d_i may fall on a non-trading day; the event day e_i is the first trading day on or after it. The return is then accumulated forward from the close of e_i over +1, +3 and +5 trading days. Only post-event days are used, so no future information enters.

Three choices fix *how* that impact is measured, and each is made for transparency as much as for rigour. The first is the *horizons*: cumulative returns are taken over short windows of +1, +3 and +5 trading days. Short windows are standard practice for daily event studies (Brown and Warner 1985) and limit the chance that unrelated, market-wide moves come to dominate the measured reaction — the longer the window, the more confounding creeps in. The second is the *baseline against which the move is read*. This work reports the *raw* cumulative return rather than a market-adjusted *abnormal* return — the return in excess of a market or expected-return model, as in a classical cumulative-abnormal-return study. Raw returns are chosen deliberately: they are directly observable and require no estimated market model that a non-expert could not scrutinise. The cost, stated plainly, is that a raw return conflates the news reaction with any market-wide move on the same days, which is precisely the confounding threat examined in Chapter 5; market adjustment is a sound refinement and is recorded as future work. The third choice is *aggregation*: the headline figure an alert shows is the *mean* impact across the retrieved precedents, a simple and interpretable summary. Because a mean can mask disagreement, the individual precedents and their impacts are always displayed alongside it, so the investor sees the spread rather than a single number in isolation.

A worked example makes the retrieve-and-reuse step concrete. The intuition is conveyed first by Figure 3.3: the embedding places each headline in a vector space where texts on the same theme fall close together, so a query lands near the precedents that resemble it and far from unrelated news.

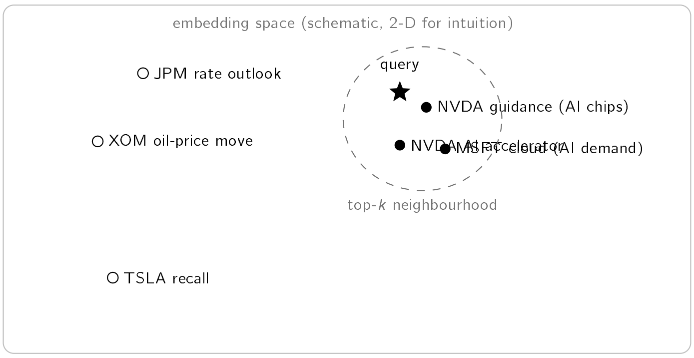

Figure 3.3: Conceptual illustration of semantic retrieval. Each headline becomes a point in a high-dimensional embedding space (drawn in two dimensions for intuition only). Texts on the same theme lie close together, so a query (star) falls near its semantically similar precedents (filled), which form the top-k neighbourhood, and far from unrelated news (hollow). The next table makes this concrete with real numbers.

Table 3.3 then carries the step through end to end on the small sample knowledge base bundled with the system. To keep the walk-through fully reproducible without downloading any model, it uses a transparent baseline encoder whose similarity reflects word overlap directly; the deployed system uses the Sentence-BERT encoder described above, whose retrieval quality is evaluated in Chapter 5. For the query *"Nvidia demand surges on AI chip orders"*, the three nearest precedents are all AI-themed and — importantly — not all of the same company: a Microsoft cloud-AI item is retrieved alongside the two Nvidia items, illustrating the *cross-ticker* generalisation on which the evaluation rests. Reusing their observed outcomes gives a mean +5-day impact of +6.5%, which the alert presents as evidence, with the explicit caveat that it records what followed comparable past news rather than a forecast.

3.3.3 Explanation Techniques

Explanation is pursued primarily through transparent design, following the interpretable-by-design argument reviewed in Section 2.3 (Barredo Arrieta et al. 2020; Rudin 2019). For every alert the system assembles three transparent ingredients: the rule or measure that

Table 3.3: Worked retrieval example on the bundled sample knowledge base. Query: *"Nvidia demand surges on AI chip orders"*; similarities and impacts are real and reproducible.

Similarity	+5-day impact	Ticker / date	Retrieved headline (excerpt)
0.60	+3.5%	NVDA, 2023-05-25	Nvidia guidance surges on soaring demand for AI data-centre chips
0.38	+10.9%	MSFT, 2023-04-25	Microsoft cloud growth accelerates on AI demand
0.38	+4.9%	NVDA, 2023-06-13	Nvidia unveils new AI accelerator to extend data-centre lead
Mean +5-day impact across the retrieved precedents: +6.5%			

produced the detection (the z-score together with its window and threshold); the retrieved historical precedents and their observed impact, shown as direct evidence; and, optionally, feature attribution over the detector using SHAP (Lundberg and S.-I. Lee 2017). Financial sentiment from a domain language model (Araci 2019) is an optional enrichment that is included only if it adds defensible value. Because each alert is rendered directly from these computed objects, the explanation is faithful to the system's actual logic by construction — a property assessed explicitly in Chapter 5.

3.4 Tools and Frameworks

The system is built on the open-source scientific-Python ecosystem, using mature, well-documented and freely available libraries for numerical computing, market-data access, sentence embeddings and figure generation. This choice is consistent with the project's constraint of using only free resources and with its emphasis on reproducibility: pinned dependencies and fixed random seeds mean the environment and the results can be recreated on another machine. The concrete engineering — the environment, the organisation of the code by component, and the automated tests — is documented at the appropriate level in Appendix A, deliberately kept out of the main text so that this chapter remains a description of *methods* rather than of software.

3.5 Responsible AI and Ethics

The work is bound by several responsible-AI commitments that follow directly from its purpose and its audience. **No advice and no prediction:** the system performs neither price prediction nor trading recommendation; it surfaces observed past outcomes as evidence and states this explicitly in every alert, so that the investor is informed rather than instructed. **Transparency:** every alert exposes its full reasoning chain, in keeping with the explainability principles of Chapter 2, so that a user can verify rather than merely trust the output. **Honest evidence:** only results that can be reproduced and explained are reported, and no data, metric or citation is fabricated. **Data stewardship:** all sources are used within their free tiers and licences, the FNSPID dataset is attributed as its Creative Commons licence requires (Dong, Fan, and Peng 2024), third-party news text is not republished, and no personal data is processed. **Security:** credentials are never stored in versioned files. Finally, the system is positioned as a decision-support aid for a non-professional investor, not as a substitute for professional advice — a limitation stated plainly rather than obscured.

3.6 Evaluation Methodology

The evaluation is defined in advance, before any experiment is run. Fixing the metrics, baselines and protocol beforehand is a deliberate guard against the temptation to choose, after the fact, the measure that flatters the result — a form of pre-registration in miniature. The full results are reported in Chapter 5; what follows is the protocol they obey.

3.6.1 Retrieval metric and relevance

The correlation engine is evaluated as an information-retrieval task (Section 2.6), using precision@k — the share of the top k retrieved precedents that are relevant. Writing Q for the set of query headlines and $R_k(q)$ for the top k items returned for query q,

$$\text{precision@}k = \frac{1}{|Q|}\sum_{q\in Q}\frac{1}{k}\sum_{d\in R_k(q)}\text{rel}(q,d), \qquad \text{rel}(q,d) = \mathbf{1}\big[\,\text{sector}(d) = \text{sector}(q)\,\big]. \tag{3.2}$$

Relevance is approximated by *sector agreement*: a precedent is counted as relevant when its company belongs to the same economic sector as the query's. This is an automatic proxy for the human judgement of "analogous", adopted because no labelled relevance set exists; its limits are stated openly with the results. Two design choices make the metric demanding rather than flattering. First, a strict *cross-ticker* constraint removes from the candidate pool every precedent that shares the query's ticker, so the system cannot score by trivially matching a company to its own past news — it must generalise *across* companies within a sector. Second, k is kept small (the headline figure uses $k = 5$), reflecting that an alert can show only a handful of precedents, so the test is whether the very top results are relevant.

3.6.2 Baselines

A single precision number means little in isolation, so semantic retrieval is compared against three baselines, each chosen to control for a specific rival explanation. The **random** baseline draws k precedents uniformly at random; its expected precision is just the base rate of same-sector items, and it therefore marks the "no-skill" floor. The **recency** baseline returns the k most recent items, testing whether simply surfacing the newest news — a tempting and cheap strategy — would do as well. The **lexical** baseline ranks by a bag-of-words, term-overlap representation under the same cosine ranking, so that the difference between it and the embedding model isolates the marginal value of *semantic* meaning over surface word matching — the central claim of the engine. Because each query sample is drawn at random, the whole comparison is repeated over five random seeds and reported as a mean with its standard deviation, so that a single lucky or unlucky draw cannot drive the conclusion; the embedding model itself is varied between two Sentence-BERT variants to show the result is a property of semantic retrieval, not of one specific model.

3.6.3 Anomaly-detection protocol

The anomaly detector is judged primarily by an argument that needs *no label*: a good universal detector should fire at a comparable rate across instruments of very different volatility, whereas a volatility-blind rule cannot. Consistency is therefore measured as the range (maximum minus minimum) of per-ticker firing rates — the smaller the range, the more uniform the detector. Only secondarily is the detector scored against an explicit *extreme-move* proxy (a daily move at or beyond the 99th percentile of that ticker's own returns), via precision, recall and F_1; because this proxy is itself volatility-relative, it is treated as supporting rather than primary evidence. An ablation over the estimation-window length completes the picture by exposing the detector's sensitivity to its one main parameter.

3.6.4 Explanation, scope, and rigour guarantees

The explanation engine is assessed for *fidelity* — whether the alert text reflects the system's actual computation, which holds by construction because each alert is rendered directly from the computed objects — while its *usefulness* to a real investor is acknowledged as requiring a human study not yet conducted, and is reported as a limitation rather than a claim. Because the system performs neither price prediction nor trading, profit-based metrics are out of scope by design, not omitted by convenience. Underpinning every experiment are three guarantees. First, **no lookahead**: any feature computed at an instant uses only information available at or before that instant, and historical impact uses only post-event windows. Second, **reproducibility**: random seeds are fixed, dependencies are pinned, and every data and results figure is produced by a versioned script. Third, **honesty**: results are reported with their caveats, and modest but genuine findings are preferred to inflated ones.

3.7 Chapter Conclusions

This chapter set out the methodology of defensible simplicity and incremental delivery, documented the two-layer data architecture in a reproducible data card, justified the models and techniques chosen for detection, retrieval, impact measurement and explanation, stated the responsible-AI commitments, and fixed the evaluation methodology and rigour guarantees in advance. The next chapter shows how these methods are integrated into CLARION, the proposed system.

Chapter 4

CLARION: An Explainable Financial-Alert System for Retail Investors

This chapter presents Clarion, the system built in this dissertation: its architecture, the historical knowledge base and correlation engine at its core, the anomaly trigger, the explanation engine that makes every alert traceable, and the way alerts reach the investor. Chapter 3 argued for the individual methods; the focus here is how they fit together into one working system.

4.1 Overview

Clarion watches the US equity market on behalf of a retail investor and raises an alert whenever one of two independent triggers fires: an abrupt price movement, or a relevant incoming news item. Each alert is accompanied by a full, traceable explanation rather than a bare notification. The system is organised as a pipeline of well-separated components, each with a single responsibility, connected by a common data schema so that the live and historical layers are directly comparable.

4.2 Architecture

The system is driven by the two triggers and converges on a single explanation step before delivery. Figure 4.1 shows the components and the flow of data between them. The live layer supplies real-time prices and news; the historical knowledge base supplies the precedents; the anomaly detector and the correlation engine produce the evidence; the explanation engine assembles a traceable account; and the delivery component sends the alert to the investor.

Three engineering principles recur throughout and make the system both testable and defensible. First, **a thin end-to-end slice was built before breadth**: the market-movement trigger was implemented and proven end to end — from price retrieval to a real alert — before the heavier correlation engine was added, reducing integration risk. Second, **pure logic is separated from input/output**, so the numerical core can be verified deterministically and offline, without a network connection or the heavy model stack. Third, **components are programmed against interfaces**: the embedding step is defined abstractly, with two interchangeable implementations — a dependency-free lexical baseline and the real Sentence-BERT model — so that one can be substituted for the other without changing anything else, which is what makes the embedding-model comparison in Chapter 5 a fair one.

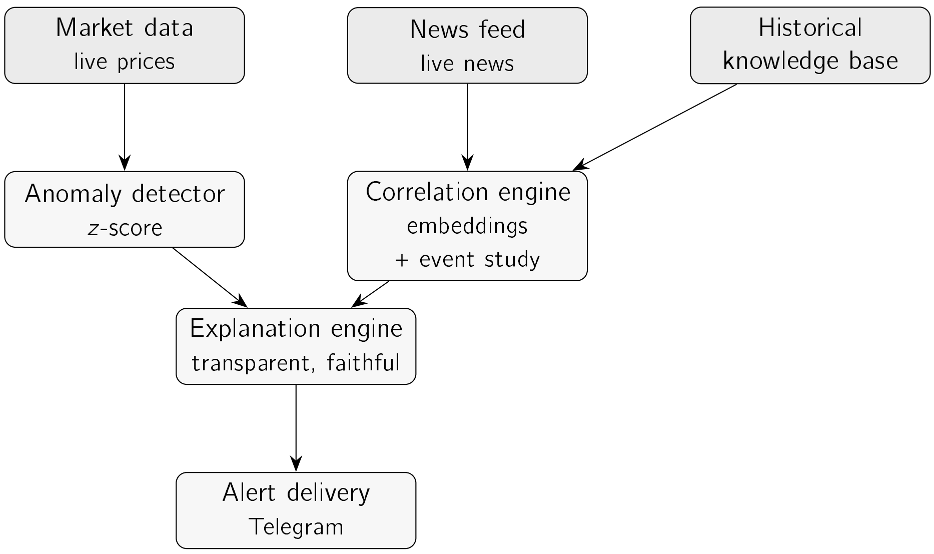

Figure 4.1: Architecture of Clarion. The live layer (market data, news feed) and the historical knowledge base feed two engines — the anomaly detector and the correlation engine — whose evidence converges on the explanation engine before an alert is delivered through Telegram. The market-movement trigger follows the left path; the news trigger the right.

Table 4.1 gathers the principal design decisions and the rationale behind each, making explicit the engineering judgement that, in a work of this kind, *is* the contribution. Every decision follows the guiding principle of defensible simplicity established in Chapter 3.

Where Figure 4.1 shows the components, Figure 4.2 traces the same system as a flow of data and processing steps: the offline knowledge-base build (top), the market trigger (middle) and the news trigger (bottom), converging on the explanation engine and the alert.

4.3 Historical Knowledge Base and Correlation Engine

The historical layer is a knowledge base of past news, each entry storing its date, ticker, headline, sentence embedding and measured post-event impact. Each news item is aligned to the market by taking the first trading day on or after the news date and measuring impact only from that day's close onward; this is the no-lookahead rule made concrete, and it avoids crediting the system with a jump already embedded in the opening price. The intended primary source is the FNSPID corpus (Dong, Fan, and Peng 2024); because building the full multi-year corpus is a long one-off job, the evaluation in Chapter 5 uses an initial knowledge base built from real recent news through the identical build process, and the full build is identified as future work.

The correlation engine has two parts. Similarity is computed with cosine over the (normalised) embeddings, returning the nearest historical items to a query. Impact is computed as the cumulative return at each horizon (+1, +3, +5 days) measured from the event close, with horizons that fall outside the available series reported as not available rather than silently dropped. Given a new item, the engine embeds it, ranks the knowledge base by similarity, and returns the top precedents with their scores — the evidence later shown to the user. Figure 4.3 traces these steps for the news trigger end to end.

Table 4.1: Principal design decisions in Clarion and their rationale.

Decision	Choice	Rationale
Detection method	Rolling z-score	Transparent; every alert is explainable to a non-expert
News representation	Sentence-BERT embeddings	Contextual meaning, yet cheap and reproducible to compare
Impact measurement	Event-study windows	Standard, reproducible; characterises an outcome, not a forecast
Explanation strategy	Transparent design + precedents	Faithful by construction, not a post-hoc approximation
Data architecture	Shared schema, two layers	Live and historical evidence are directly comparable
Component coupling	Programmed to interfaces	Testable offline; enables fair model substitution
Delivery channel	Telegram bot	Free, ubiquitous, no bespoke client needed

Where Figure 4.3 shows *what* transforms the data, Figure 4.4 shows the same trigger as an interaction *over time*: which component calls which, in what order, and what each one returns. Reading top to bottom, the trigger polls the live feed; a new headline is handed to the correlation engine, which embeds it and queries the knowledge base; the base returns the nearest precedents with their measured impact; and the engine passes that evidence to the explanation engine, which renders and delivers the alert. The dashed arrows are returns, making explicit that the knowledge base is consulted but never decides — the reasoning stays in the open.

4.4 Anomaly Trigger

The anomaly trigger applies Equation 3.1 directly. For the day under evaluation it computes the mean and standard deviation over the window of strictly preceding days, forms the z-score of the latest return, and returns the decision together with every quantity that produced it — the z-score, window, threshold, mean and standard deviation. Exposing these values is what allows the explanation engine to be faithful by construction, and the evaluation applies exactly this rule across entire price series so that the experiments use the same logic as production. Figure 4.5 traces the market trigger from a daily price update to a delivered alert.

As with the news trigger, the same flow can be read as an interaction over time (Figure 4.6): the scheduler requests prices, the detector standardises the latest return against its rolling window, and — only when the absolute z-score exceeds the threshold — it hands the move together with every quantity behind it to the explanation engine for delivery.

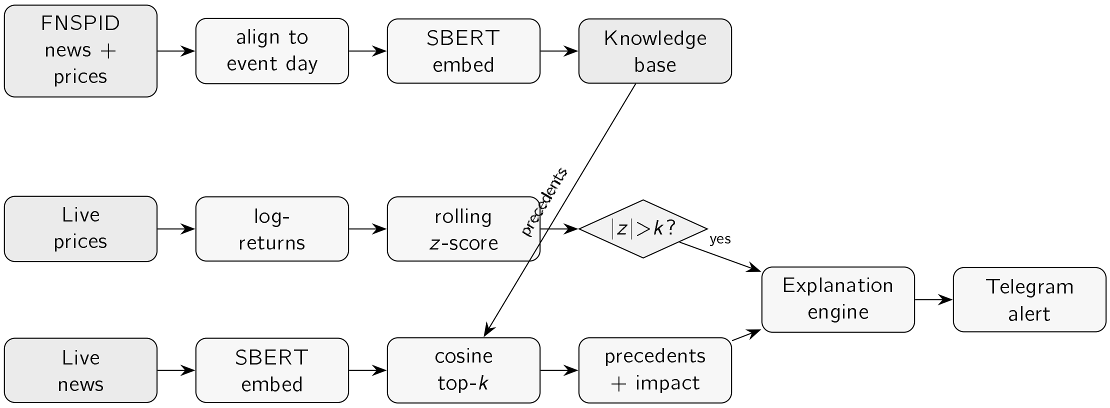

Figure 4.2: The end-to-end flow of Clarion. Offline (top), FNSPID news is aligned to prices and embedded into the knowledge base. Online, the market trigger (middle) turns live prices into a z-score and fires when it exceeds the threshold; the news trigger (bottom) embeds an incoming headline, finds its nearest precedents in the knowledge base, and attaches their observed impact. Both converge on the explanation engine, which delivers a single explained alert.

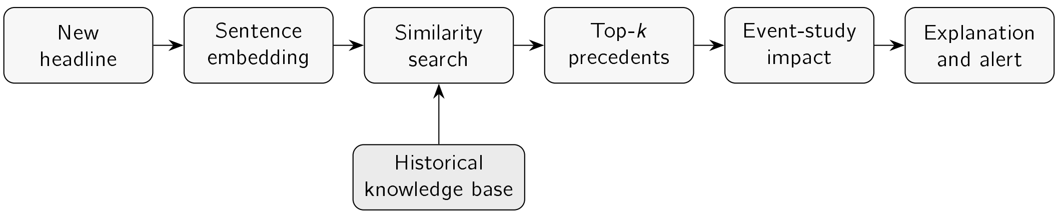

Figure 4.3: The news trigger from an incoming headline to a delivered alert. The headline is embedded, matched against the historical knowledge base by similarity, and its nearest precedents are returned; their post-event impact is measured and assembled, with the headline itself, into a single explained alert.

4.5 Explanation Engine

The explanation engine renders alerts directly from the objects produced upstream, which guarantees that the text never diverges from the computation. For the market trigger, it states the move, the z-score, the threshold and window, and the recent mean and standard deviation. For the news trigger, it lists the retrieved precedents (date, ticker, similarity score, measured impact and headline) and the average impact across them, and it always ends with an explicit disclaimer that the figures are observed past outcomes, not a prediction — keeping the system within its no-forecasting scope.

4.6 Alert Delivery

Alerts are delivered to the investor through the Telegram messaging platform, chosen because it offers a free, widely used bot interface. The two end-to-end flows — retrieve, detect, explain and send for the market trigger; embed, retrieve precedents, explain and send for the news trigger — each encapsulate a complete run; periodic scheduling and deployment are deliberately left outside the present scope. A real alert was successfully delivered during

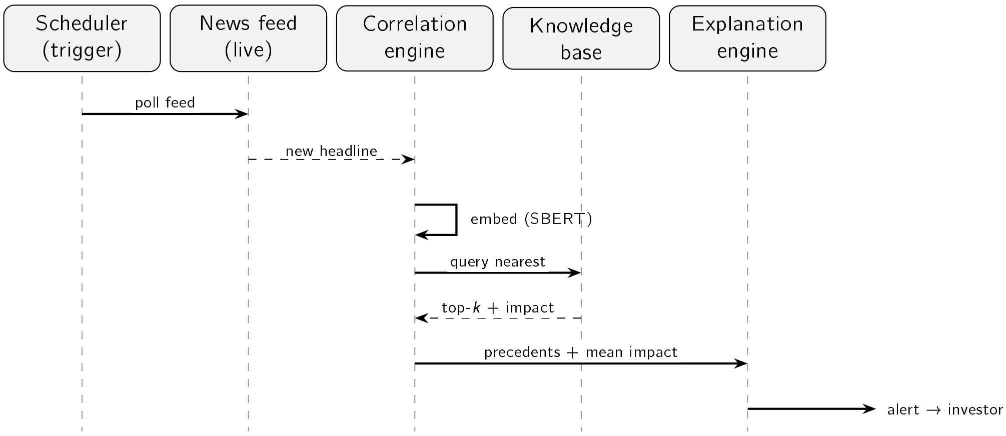

Figure 4.4: The news trigger as an interaction over time. Solid arrows are calls, dashed arrows are returns; time runs downward. The correlation engine orchestrates the work — embedding the headline, consulting the knowledge base, and assembling the retrieved precedents and their measured impact — before the explanation engine renders and delivers the alert. This complements the data-flow view of Figure 4.3.

Figure 4.5: The market-movement trigger. The latest daily return is standardised against the rolling mean and standard deviation of the preceding window; if the absolute z-score exceeds the threshold k, the move and all the quantities behind it are passed to the explanation engine and delivered.

testing; Figure 4.7 reproduces the format the investor receives, with the full reasoning chain visible in a single message.

4.7 Deployment and Practical Use

CLARION runs as two entry points, one per trigger, each of which performs a complete cycle. The market trigger fetches recent prices, computes returns, detects, explains and sends; the news trigger takes a headline, embeds it, retrieves precedents, explains and sends. In day-to-day use these would be invoked on a schedule — the market trigger once per trading day after the close, the news trigger whenever a watched feed publishes — but the scheduler and the hosting are deliberately outside the project's scope. The entry points already encapsulate everything such a scheduler would call.

Running the system needs three things, all free: a market-data source for prices, a company-news or RSS source for headlines (an API key, kept out of version control), and the historical knowledge base, built once from the corpus. Appendix A gives the exact steps. A real alert was delivered to a Telegram account during testing, so the path from raw data to the investor's phone is proven end to end, not just designed.

Two practical limits are worth stating plainly. First, the free news tier returns only a recent window, so the knowledge base is shallower than the multi-year build it is designed for; this affects coverage, not the method. Second, CLARION is a decision-support prototype rather

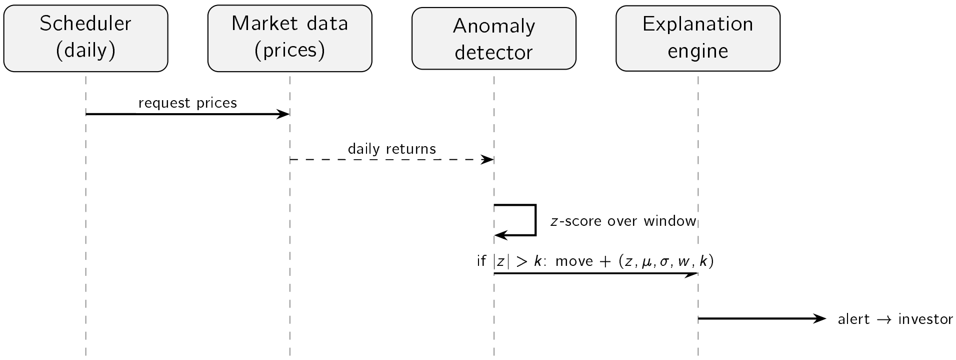

Figure 4.6: The market-movement trigger as an interaction over time. Solid arrows are calls, dashed arrows returns; time runs downward. The detector standardises the latest return against its rolling window and forwards the move — with the z-score, window, threshold, mean and standard deviation — to the explanation engine only when the threshold is exceeded. This complements the data-flow view of Figure 4.5.

than a production service: it has no user accounts, no persistence beyond the knowledge-base file, and no automatic recovery from a provider outage. None of these is a research limitation; each is a known, ordinary step on the road from prototype to product, and is recorded here for honesty rather than hidden.

4.8 Chapter Conclusions

This chapter described CLARION at the level of its architecture and design: a two-trigger, two-layer pipeline whose components are separated by responsibility and connected by a common schema, with explainability built in by rendering every alert directly from the system's own computation. The next chapter evaluates the system's two core components and its end-to-end behaviour on real data.

CLARION — financial alerts bot

News alert — NVDA

"Nvidia raises guidance on surging demand for AI data-centre accelerators"

Potential impact (from 5 similar past events): average 1-day move −2.0%

Historical precedents:
• 2026-06-24 MSFT (sim 0.68) +1d −3.5%
• 2026-06-24 META (sim 0.68) +1d −2.7%
• 2026-06-24 GOOGL (sim 0.64) +1d −0.5%

Precedents are retrieved by semantic similarity; the impact is the observed past outcome, not a price prediction.

Figure 4.7: A CLARION news alert as delivered through Telegram. A single message carries the full, traceable reasoning chain: the detected event, the average observed impact of analogous past events, the individual precedents with their similarity scores and measured impact, and an explicit no-prediction disclaimer.

Chapter 5

Case Studies

This chapter evaluates CLARION through three case studies on real data. The first looks at the anomaly trigger across stocks of very different volatility; the second at the correlation engine and the precedents it retrieves; the third walks through a complete alert and asks whether its explanation is faithful. Every number below comes from a versioned script with a fixed seed, the evaluation followed the plan set out in Chapter 3, and each study states its own limitations.

5.1 Common Experimental Setup

Two real datasets are used. For the anomaly trigger, daily closing prices for the fifteen large-capitalisation tickers listed in the data card were retrieved from a free market-data service over a fixed three-year window (June 2023 to June 2026), yielding 750 daily log-returns per ticker. For the correlation engine, 3,714 real financial-news headlines for the same tickers were collected through a free company-news service — the most recent months available on the free tier — spanning five sectors (technology, banking, energy, health and consumer staples). News headlines and prices are normalised to a common schema so that the live and historical layers are directly comparable. This recent window of a few months is enough for the retrieval experiment; a full impact analysis still needs the multi-year FNSPID corpus, left as future work. The no-lookahead, reproducibility and honesty guarantees of Section 3.6 apply to every study below.

5.2 Case Study 1: Transparent Anomaly Detection on US Equities

The question is whether the rolling z-score (Chandola, Banerjee, and Kumar 2009) is a better universal detector of abrupt moves than a naïve fixed-percentage threshold. The proxy label for a "true" anomaly is a daily move in the upper percentile of *that ticker*'s own return distribution ($|r| \geq$ the 99th percentile); the baseline is a single global threshold ($|r| \geq 3\%$) applied to every ticker. This label is itself volatility-relative, so any agreement score it produces is only supporting evidence; the primary argument, made first below, is the firing-rate consistency, which needs no label at all.

Firing-rate consistency (main argument). The strongest evidence is independent of any label. A good universal detector should fire at a comparable rate across instruments; a volatility-blind threshold cannot. Table 5.1 and Figure 5.1 show that the fixed threshold fires on roughly 1% of days for a calm stock (KO) but on about 35% of days for a volatile one (TSLA, NVDA), whereas the z-score fires at a near-constant rate ($\approx$ 2–3%) for all tickers. The range of firing rates across tickers is more than twenty times tighter for the z-score (0.015 versus 0.344), confirming that normalising by recent volatility makes detection comparable across the market.

Table 5.1: Firing rate across the fifteen tickers (three years of daily data).

Method	Min rate	Max rate	Range
z-score (window 20, ± 3)	0.016	0.031	**0.015**
Fixed threshold ($\lvert r \rvert \geq 3\%$)	0.009	0.353	**0.344**

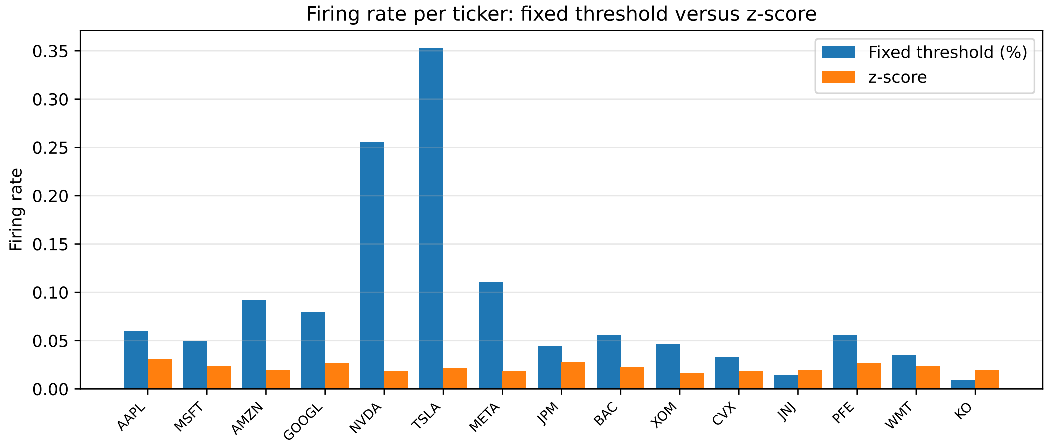

Figure 5.1: Firing rate per ticker. The fixed percentage threshold varies dramatically with each stock's volatility, while the z-score is stable across all tickers.

A worked example. Figure 5.2 shows the detector at work on a single volatile stock (Tesla) over the three-year window. The flagged days are not simply the largest raw moves: they are the moves that are large *relative to the asset's own recent volatility*, which is exactly the behaviour a fixed percentage threshold cannot reproduce. Over this period the detector flags sixteen of the seven hundred and fifty trading days (about 2%), consistent with the near-constant firing rate reported above.

To make this concrete, consider the single largest surprise in the window. On 24 October 2024, following its quarterly earnings, Tesla's daily log-return was $+0.198$ — a price move of about $+22\%$. Measured against the preceding twenty days, whose mean log-return was -0.92% and standard deviation 2.73%, this is a z-score of $+7.6$, far beyond the threshold $k = 3$ (Table 5.2). The same rule that leaves an ordinary two-percent wobble unflagged identifies this earnings reaction at once, and exposes precisely the quantities that justify the alert — the move, the recent norm it is judged against, and the resulting z-score — which is what the explanation engine then shows the investor.

Agreement with the proxy label (supporting). Against the per-ticker extreme-move label, the z-score attains an F_1 of 0.516 (precision 0.381, recall 0.800) versus 0.218 for the fixed threshold (precision 0.122, recall 0.992): the fixed rule catches almost every large raw move but at very low precision. A window ablation (Figure 5.3) shows F_1 rising with

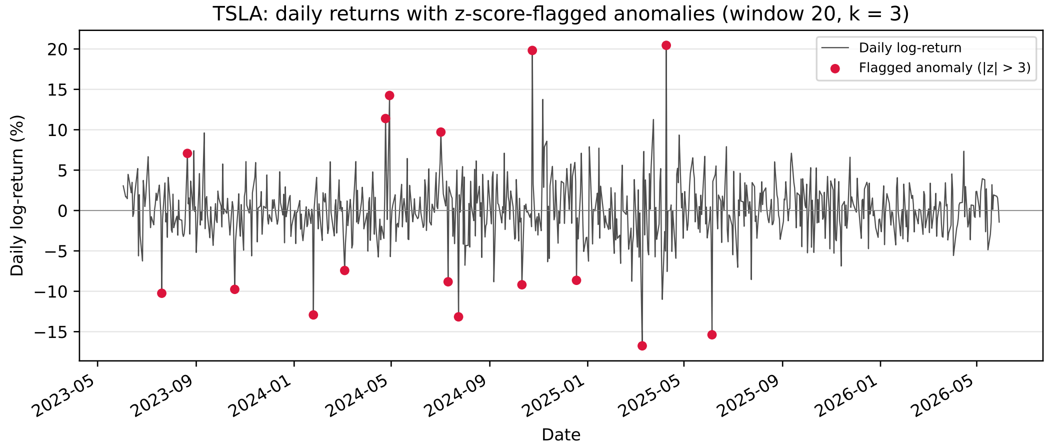

Figure 5.2: Daily log-returns of a volatile stock (Tesla) over the evaluation window, with the days flagged by the z-score detector (window 20, $k = 3$) marked. The detector responds to volatility-relative surprise rather than to absolute move size.

Table 5.2: Worked anomaly: Tesla's post-earnings move on 24 October 2024 (window 20, $k = 3$); real, reproducible figures.

Quantity	**Value**
Trailing-20-day mean of log-returns, μ	-0.92%
Trailing-20-day standard deviation, σ	2.73%
Day's log-return, r (a price move of $\approx +22\%$)	$+19.82\%$
z-score, $(r - \mu)/\sigma$	$+\mathbf{7.61}$
Flagged at $k = 3$?	yes

the estimation window (0.385, 0.516 and 0.678 for 10, 20 and 60 days), indicating that a longer volatility estimate yields a steadier baseline before it begins to saturate. This label is itself volatility-relative, so some circularity is unavoidable; that is precisely why the firing-rate consistency above — which depends on no label — is treated as the primary evidence.

5.3 Case Study 2: News–Market Precedent Retrieval

The central claim of this work is that semantic retrieval finds genuinely analogous precedents. This is measured as precision@k with a sector proxy, in a *cross-ticker* setting: for each query the k most similar headlines *from other companies* are retrieved, and precision is the fraction sharing the query's sector. Excluding the query's own company prevents the trivial solution of matching on the company name and tests thematic analogy instead. It is worth being clear about what this measures: same sector is an automatic stand-in for relevance, so the metric shows whether retrieved news is thematically related, not whether a person would judge it a useful precedent. That gap is real, and a human relevance study is left as future work; the proxy's merit is that it is objective and reproducible, which is what a first evaluation needs. To check that the result is not tied to one model, *two* Sentence-BERT models (Reimers and Gurevych 2019) — the lightweight default, MiniLM, and the larger MPNet — are compared against a lexical (hashing) baseline and two trivial baselines: random retrieval (the sector

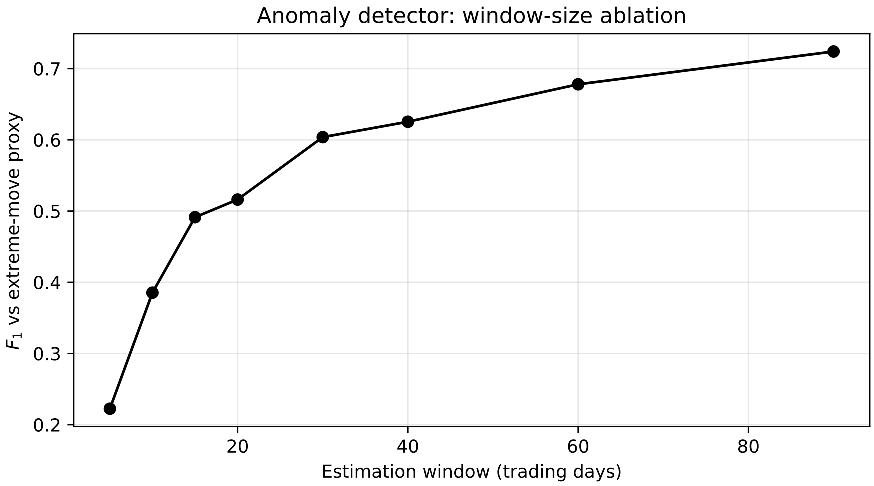

Figure 5.3: Effect of the estimation-window length on the detector's F_1 against the extreme-move proxy. A longer window yields a steadier volatility baseline and higher agreement, with diminishing returns at the longest windows.

base rate) and recency. To guard against sampling artefacts, five hundred queries were sampled in each of five independent runs (seeds 42–46); Table 5.3 reports the mean and standard deviation.

Table 5.3: Cross-ticker precision@k by sector, mean $\pm$ std over five runs (3,714 real headlines, five sectors).

Method	**P@5**	**P@10**
SBERT — MiniLM (default)	0.514 ± 0.015	0.478 ± 0.011
SBERT — MPNet	$\mathbf{0.538 \pm 0.011}$	$\mathbf{0.506 \pm 0.010}$
Lexical baseline (hashing)	0.346 ± 0.011	0.314 ± 0.008
Recency	0.126 ± 0.006	0.106 ± 0.005
Random (base rate)	0.240 ± 0.004	0.240 ± 0.004

As Table 5.3 and Figure 5.4 show, the default MiniLM model reaches a precision@5 of 0.514 — about 2.1 times the random base rate (0.240) and well above the lexical baseline (0.346). The stronger MPNet model does slightly better (0.538), which suggests the advantage belongs to semantic embeddings in general rather than to one particular model; the small standard deviations (around 0.01) show the gap is robust rather than a sampling artefact. Recency performs worst, below even the random base rate, so simply surfacing the latest news is no substitute for semantic matching. The lift over both the random and lexical baselines is the honest measure of the value added by the embeddings.

Impact measurement. For each retrieved precedent the engine reports the observed post-event return at $+1$, $+3$ and $+5$ trading days, measured from the event-day close in the style of an event study (Brown and Warner 1985). These figures are observed outcomes presented as evidence, never a prediction. On the worked knowledge base the measured impacts are directionally coherent — for example, the competitive-threat news cluster in the end-to-end case study below is followed by uniformly negative one-day returns — which provides face

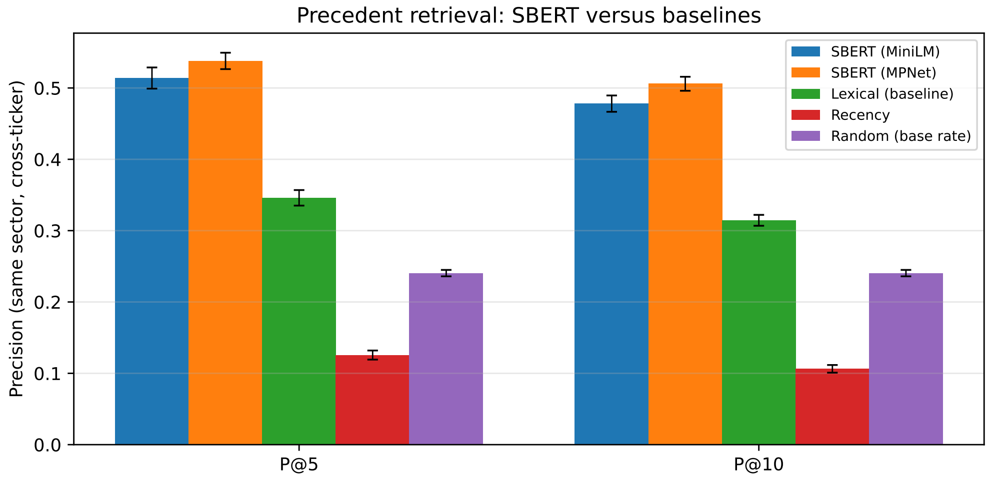

Figure 5.4: Cross-ticker precision@k by sector. SBERT clearly exceeds the lexical, recency and random baselines; error bars show ± 1 standard deviation over the five runs.

validity for the measurement; a rigorous multi-year impact study on the full FNSPID corpus is identified as future work.

Retrieval by sector. The aggregate figure hides useful variation across sectors. Table 5.4 and Figure 5.5 break the retrieval down by the query's sector, using the whole population of each sector as queries (so the figures are exact rather than sampled). Two readings matter. The raw precision@5 is highest for technology (0.712), but largely because technology dominates the corpus (47% of headlines) and therefore has a high random base rate (0.429); the fair cross-sector measure is the *lift* over that base rate. By that measure the engine adds the most value in **energy** (+0.377) and **health** (+0.348) — sectors whose vocabulary is distinctive (production, output, barrels; trial, drug, approval) — and the least in **consumer** (+0.100), whose headlines are more generic and thematically overlap with other sectors. Retrieval is positive in every sector, but its benefit is greatest where the domain language is most specific.

Table 5.4: Cross-ticker precision@k per sector (whole-population queries, MiniLM).

Sector	**N**	**P@5**	**P@10**	**Random**	**Lift P@5**
Technology	1736	0.712	0.675	0.429	+0.283
Energy	497	0.448	0.408	0.072	**+0.377**
Health	494	0.419	0.379	0.071	+0.348
Banking	496	0.272	0.228	0.072	+0.200
Consumer	491	0.171	0.150	0.071	+0.100

Why some sectors retrieve better: a qualitative look. The per-sector gap has a concrete explanation, visible in individual queries. In technology, different companies genuinely share themes: a headline about NVIDIA's data-centre accelerators retrieves, from the Microsoft, Meta and Alphabet feeds, articles about the same wave of AI-chip competition

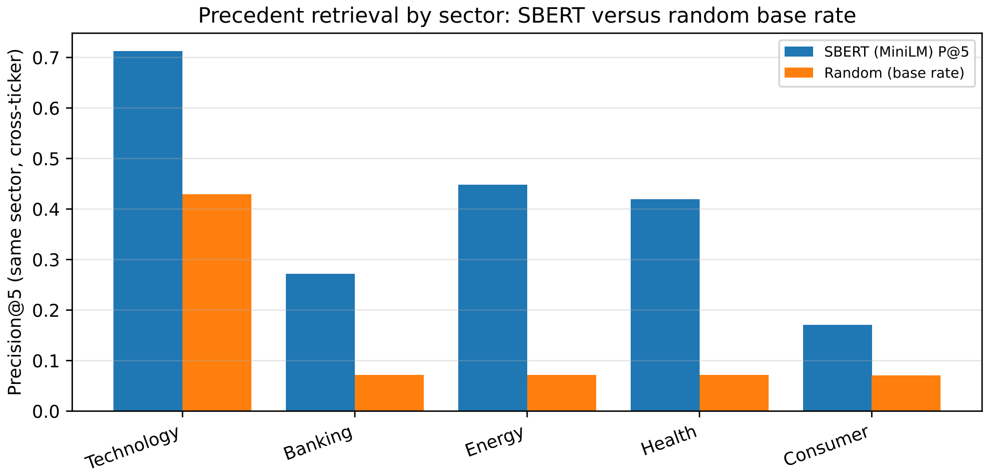

Figure 5.5: Precision@5 per sector against the random base rate. Retrieval exceeds the base rate in every sector; the gap (lift) is largest for energy and health, whose vocabulary is most distinctive, and smallest for the broad consumer sector.

(Case Study 3), so cross-ticker precision is high. The consumer sector behaves differently. A query about Walmart's grocery guidance retrieves mostly Walmart's own items plus a macro "US retail sales" story that happens to be carried on bank feeds — adjacent in theme, but not a consumer analogue from another consumer company. A query about Coca-Cola retrieves almost only other Coca-Cola items, about its pricing power and beverage volumes, which simply do not resemble Walmart's store-traffic news. The shared label "consumer" is too broad: its members' news is idiosyncratic, so cross-ticker analogues are scarce and the lift over the base rate is small. This is exactly the pattern the numbers show, and it points to a clear refinement for future work: grouping by finer-grained themes rather than by sector.

5.4 Case Study 3: End-to-End Alert and Explanation Faithfulness

To illustrate the news trigger end to end, consider a new headline for NVIDIA, *"Nvidia raises guidance on surging demand for AI data-centre accelerators"*. The system embeds it, retrieves the most similar precedents from the knowledge base built from the real news corpus, and renders the alert below (top five precedents, one-day horizon):

```
News alert for NVDA
"Nvidia raises guidance on surging demand for AI data-centre
accelerators"
Potential impact (from 5 similar past events):  average 1-day move:
−1.97%
Historical precedents:
- 2026-06-24 MSFT (sim 0.68) +1d −3.46%:  "Qualcomm debuts line of AI
data center chips..."
- 2026-06-24 NVDA (sim 0.68) +1d −1.64%:  "Qualcomm debuts line of AI
data center chips..."
```

```
- 2026-06-24 META (sim 0.68) +1d −2.65%:  "Qualcomm debuts line of AI
data center chips..."
- 2026-06-24 GOOGL (sim 0.64) +1d −0.46%:  "This AI Infrastructure
Stock Could Be Bigger Than Nvidia..."
- 2026-06-24 NVDA (sim 0.58) +1d −1.64%:  "Tecan Accelerates
Data-Driven Lab Journey...  NVIDIA"
Note:  precedents are retrieved by semantic similarity; the impact is
the observed past outcome, not a price prediction.
```

Every retrieved precedent concerns AI data-centre chips and competition for NVIDIA, even though the articles were filed under different company feeds (Microsoft, Meta, Alphabet and NVIDIA itself). Strikingly, the same story — a rival entering the AI data-centre chip market — surfaces across three different company feeds, which the engine correctly groups as one theme. This is direct evidence that retrieval is driven by *meaning* rather than the company name or exact keywords — the very behaviour the correlation engine is designed for. In this instance the precedents happen to be uniformly negative (a competitive-threat cluster), and their average is reported as an observed past outcome, not a prediction. The alert ends with an explicit disclaimer, keeping the system within its no-forecasting scope.

Explanation faithfulness. Faithfulness is guaranteed by construction: the alert text is rendered directly from the same objects the system computes — the z-score, window and threshold for the anomaly path, and the exact retrieved precedents with their similarity scores and measured impacts for the news path — so the explanation cannot diverge from the underlying computation. This is checked programmatically by an automated test that asserts the rendered alert reproduces exactly each retrieved precedent's date, ticker and similarity score and introduces none that were not retrieved; it is the kind of transparent, design-time explainability advocated in the XAI literature (Barredo Arrieta et al. 2020). Usefulness to a retail investor is inherently subjective; a small rubric-based human assessment (clarity, completeness, actionability) is planned but has not yet been conducted, and is therefore reported as a limitation rather than a result.

5.5 Discussion and Threats to Validity

The three case studies support the core design: volatility-normalised detection is consistent across the market where a fixed threshold is not, semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines, and the end-to-end alert is faithful to its own computation. All results rest on real data and are reproducible from versioned scripts.

The results also have clear limits, which are set out below under the four standard categories of validity so that nothing is glossed over.

Construct validity — do the metrics measure what is claimed? Two proxies carry weight here. Retrieval relevance is judged by a *sector* label, an automatic stand-in for the human judgement of "analogous", so a same-sector precedent on an unrelated topic counts as a hit; the qualitative inspection in Section 5.3 shows this proxy is reasonable but imperfect. Likewise, the anomaly "extreme move" label is volatility-relative, so it is not an independent ground truth — which is precisely why the primary anomaly argument is the label-free *firing-rate consistency*, not agreement with that label. Finally, the event-study figure measures the price change that *followed* a news item, which is an outcome, not a causal effect of the news; it is reported as evidence, never as a forecast.

Internal validity — could something else explain the effect? The retrieval advantage might, in principle, be inflated by trivially matching a company's news to its own past news; this is removed by construction, because the evaluation forbids same-ticker precedents and measures *cross-ticker* precision only. Lookahead is excluded throughout: every quantity uses information available at the time, and impacts are accumulated strictly forward from the event-day close. The main residual threat is confounding — a price move in a post-event window may be driven by market-wide factors rather than the news — which the short windows (+1, +3, +5 days) limit but do not eliminate, and which the no-causation framing acknowledges openly.

External validity — how far do the findings generalise? The study covers fifteen large-capitalisation US tickers across five sectors over a recent window, so the numbers need not transfer to small-capitalisation stocks, to other markets, or to other periods, and news coverage is thinner outside large caps. The corpus is a recent window from a free service rather than the multi-year FNSPID history; the full corpus is identified as the natural way to broaden this scope.

Statistical-conclusion validity — are the comparisons sound? The retrieval results are averaged over five random samples with standard deviations reported (the error bars in Figure 5.4), and the gap between semantic retrieval and every baseline is large relative to that spread; still, five runs on one corpus is a modest sample, and no formal significance test is claimed. The anomaly comparison is reported descriptively, as a difference in firing rates across instruments, rather than as a hypothesis test.

One limit is deliberate rather than a weakness: the system makes no price prediction and evaluates no trading profit, so those questions are out of scope by design. The full FNSPID corpus, a human relevance and usefulness study, and a multi-year impact analysis are the natural next steps, and are carried forward to the conclusions.

5.6 Chapter Conclusions

Across three case studies on real data, CLARION's anomaly trigger fired far more consistently than a fixed threshold, its correlation engine retrieved sector-analogous precedents well above every baseline and independently of the embedding model, and its explanations were faithful to the underlying computation by construction. The next chapter draws these findings together against the research questions.

Chapter 6

Conclusions

This dissertation set out to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers and delivered through Telegram. This chapter summarises what was achieved against the research questions, revisits the contributions, and states the limitations and the directions they open.

6.1 Main Conclusions

The work is best summarised by returning to the three research questions of Chapter 1.

RQ1 — transparent detection of abrupt movements. The answer is yes. A rolling z-score detector flags abnormal returns and, more importantly, does so at a steady rate *across stocks of very different volatility*, where a fixed-percentage threshold cannot. As reported in Chapter 5, the firing rate varied by only 0.015 across the fifteen tickers for the z-score versus 0.344 for the fixed baseline, and the z-score reached an F_1 of 0.516 against an extreme-move proxy, more than double the baseline. The method is simple enough that every alert can be explained to the investor.

RQ2 — analogous precedents without lookahead. The answer is affirmative for retrieval. Semantic retrieval with Sentence-BERT recovered markedly more sector-analogous precedents than every trivial alternative — a cross-ticker precision@5 of 0.514 ± 0.015 (over five runs) against 0.346 for a lexical baseline, 0.240 for random and 0.126 for recency — and a per-sector breakdown showed the lift is greatest where domain vocabulary is most distinctive (energy, health). The end-to-end case study showed it surfacing genuinely topical precedents across different company feeds. Impact is measured strictly after the event, in the style of an event study, so no future information leaks into the evidence. The measurement machinery is in place and behaves coherently; a large-scale, multi-year validation of impact magnitudes remains for future work.

RQ3 — faithful and useful explanations. Faithfulness is achieved: because each alert is rendered directly from the same objects the system computes (the z-score and its parameters, or the retrieved precedents with their scores and measured impacts), the explanation cannot diverge from the underlying logic, and every alert closes with an explicit no-prediction disclaimer. Usefulness to a retail investor is plausible by design but has not yet been measured with a human study, and is therefore reported as an open point rather than a settled result.

6.2 Research Questions and Objectives Achieved

The general objective — to design, implement and evaluate an explainable, reproducible alerting system driven by two triggers — was met: CLARION is a working system whose two triggers were proven end to end and whose two core components were evaluated on real data. The supporting objectives were also met: a thin end-to-end slice was delivered first; each component was kept in its simplest defensible form; the evaluation followed a design fixed in advance; and the whole system is reproducible from versioned scripts. RQ1 and RQ2 are answered affirmatively on real data, while RQ3 is answered for faithfulness and left partly open for usefulness, pending a human study.

6.3 Contributions Revisited

As an Artificial Intelligence *Engineering* dissertation, the contribution is the integration, application and critical evaluation of existing components into a working, explainable and reproducible whole, rather than a new algorithm. Three concrete contributions stand. First, a documented and reproducible methodology for **news–market impact correlation** that combines sentence-embedding retrieval with event-study impact windows, with explicit choices of similarity metric and horizon and explicit avoidance of lookahead. Second, **CLARION**, an **XAI-first alerting system** whose entire reasoning chain — detection, evidence, sources and precedents — is exposed to a non-expert user and is faithful by construction. Third, a **critical, honest evaluation** of each core component against trivial and lexical baselines on real data, with results, ablations and limitations reported transparently and reproducibly from versioned scripts.

6.4 Limitations

Several limitations remain. The evaluation used a recent, few-month news corpus from a free service rather than the multi-year FNSPID history, which bounds the impact analysis and makes the retrieval figures, though real, preliminary. The relevance proxy is sector agreement, an automatic stand-in for human judgement, and the anomaly proxy label is volatility-relative (hence the label-free firing-rate argument is given primacy). News is represented by short headlines only, which limits the semantics that can be captured. The retrieval result is averaged over five runs, but no human usefulness study has yet been conducted. Finally, and by design, the system performs no price prediction or trading, so profit-based questions are out of scope rather than unanswered.

6.5 Future Work

The limitations map directly onto future work. The most immediate step is to build the full multi-year knowledge base from FNSPID, for which the streaming pipeline is already in place, enabling a richer impact analysis with dispersion and direction-consistency statistics. A small human study, applying a clarity–completeness–actionability rubric to a set of alerts, would address the open question of usefulness (RQ3). The retrieval could be strengthened by embedding full article bodies rather than headlines and by comparing further embedding models, and the explanation could be optionally enriched with financial sentiment or feature attribution where it adds value. Together these would turn a sound prototype into a more thoroughly validated system, without giving up the simplicity and transparency that make it defensible in the first place.

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Appendix A

Reproducibility

This appendix records the engineering detail needed to reproduce the system and its results, kept out of the main text so that the body reads as a dissertation rather than a software specification.

A.1 Environment

The system is implemented in Python 3.12, chosen for the maturity of its packages for the machine-learning stack. Dependencies are pinned and frozen in a lock file so the environment can be recreated across machines. The numerical and data handling rely on established open-source libraries; sentence embeddings use the Sentence-BERT framework on a CPU build of the underlying deep-learning library, installed from a dedicated index to avoid the much larger GPU download; figures are produced with a standard plotting library.

A.2 Repository Organisation

The code is organised as one package per component, mirroring the architecture of Figure 4.1: market data, news retrieval, anomaly detection, the correlation engine (similarity and event study), the historical knowledge base, the explanation engine, and alert delivery, with a small orchestration layer wiring them into the two end-to-end flows. Pure logic is separated from input/output, and heavy or networked dependencies are loaded lazily, so the numerical core is unit-tested without a network connection or the model stack. Secrets are read only from a local, ignored environment file and are never written to versioned files.

A.3 Tests and Reproducible Results

The system is covered by an automated test suite spanning the anomaly detector, the event study, similarity, the knowledge base, the news parser, the explanation engine and the evaluation modules, together with an offline end-to-end smoke test of the news trigger. Tests that require network access or heavy models are marked and excluded from the default run so that it is fast, deterministic and offline. All experimental results in Chapter 5 are regenerated by scripts with a fixed random seed, and the result figures are written directly into the thesis figure directory, closing the loop between code and document.

A.4 Data and Attribution

Large data and models are never versioned; they are recreated by the data scripts, and only small samples are committed. The FNSPID dataset (Dong, Fan, and Peng 2024) is used under its CC BY-SA 4.0 licence, with attribution as required; the live market-data and news services are used within their free tiers.